

05 – Database Administration

Give access privileges to users on your Database

Jérôme Landré – jerome.landre@ju.se

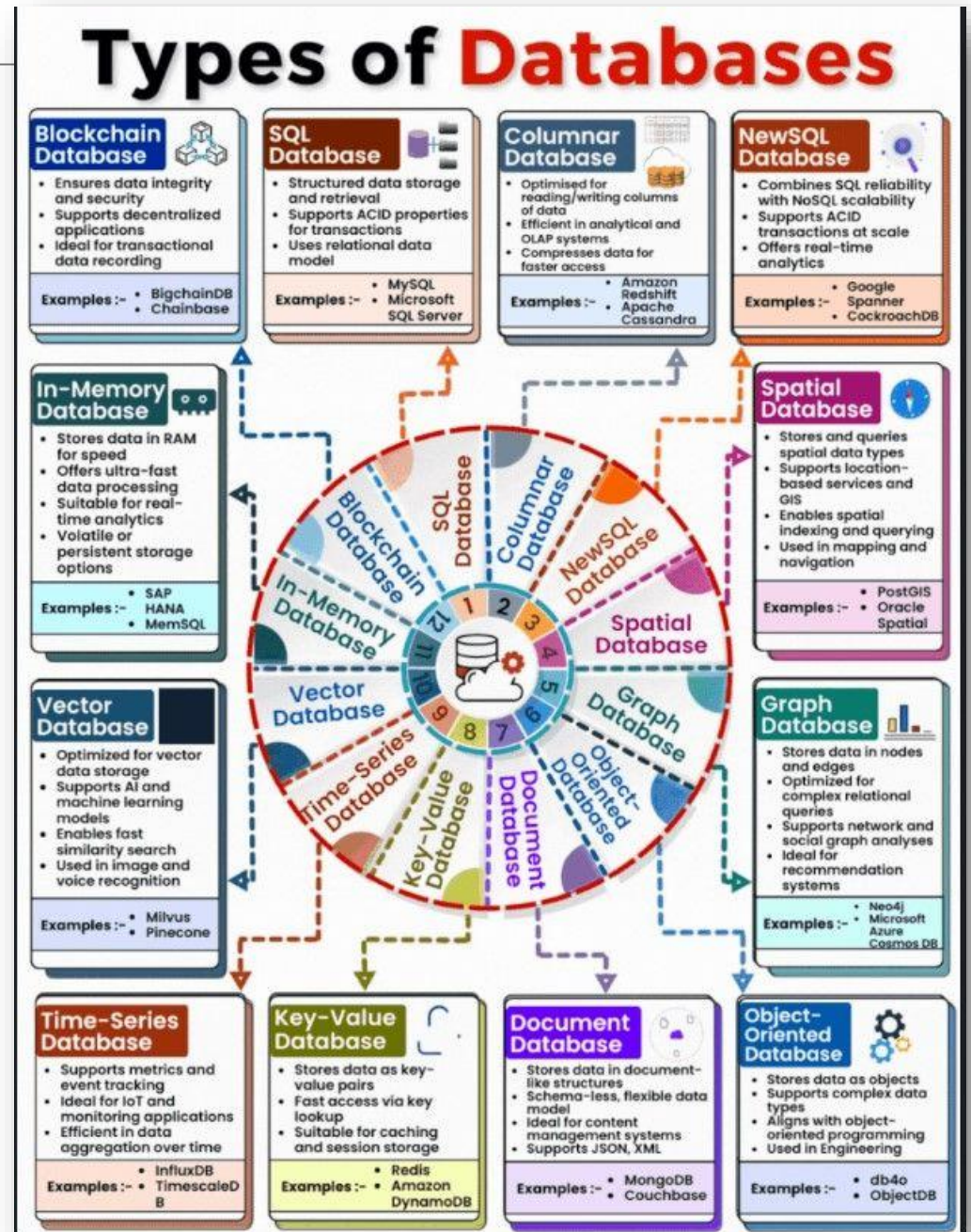
OUTLINE

1. Database Types
2. Database Rankings
3. Database Architecture
4. Secure Database Architecture
5. Keys to security
6. Permissions in MS SQL Server
7. Backups in MS SQL Server
8. Permissions in MySQL (and MariaDB)
9. Backups in MySQL (and MariaDB)



1. Database Types

- There are a lot of database types available for developers
- Depending on the application, developers will choose among them



2. Database Rankings

GAME: SQL or NoSQL

1. Oracle
2. MySQL
3. Microsoft SQL Server
4. PostgreSQL
5. MongoDB
6. Snowflake
7. DataBricks
8. Redis
9. IBM Db2
10. Apache Cassandra

2. Database Rankings

Rank			DBMS	Database Model	Score		
Apr 2026	Mar 2026	Apr 2025			Apr 2026	Mar 2026	Apr 2025
1.	1.	1.	Oracle	Relational, Multi-model 	1157.93	-24.53	-73.12
2.	2.	2.	MySQL	Relational, Multi-model 	857.69	-0.65	-129.41
3.	3.	3.	Microsoft SQL Server	Relational, Multi-model 	702.08	-9.40	-82.93
4.	4.	4.	PostgreSQL	Relational, Multi-model 	681.35	+1.27	+14.10
5.	5.	5.	MongoDB	Document, Multi-model 	385.02	+1.44	-15.03
6.	6.	6.	Snowflake	Relational	217.38	+6.13	+49.28
7.	7.	↑ 12.	Databricks	Multi-model 	150.47	+4.67	+51.19
8.	8.	↓ 7.	Redis	Key-value, Multi-model 	146.53	+1.35	-7.58
9.	9.	9.	IBM Db2	Relational, Multi-model 	113.25	+1.87	-12.79
10.	↑ 11.	↑ 11.	Apache Cassandra	Wide column, Multi-model 	102.62	+0.74	-5.59

2. Database Rankings

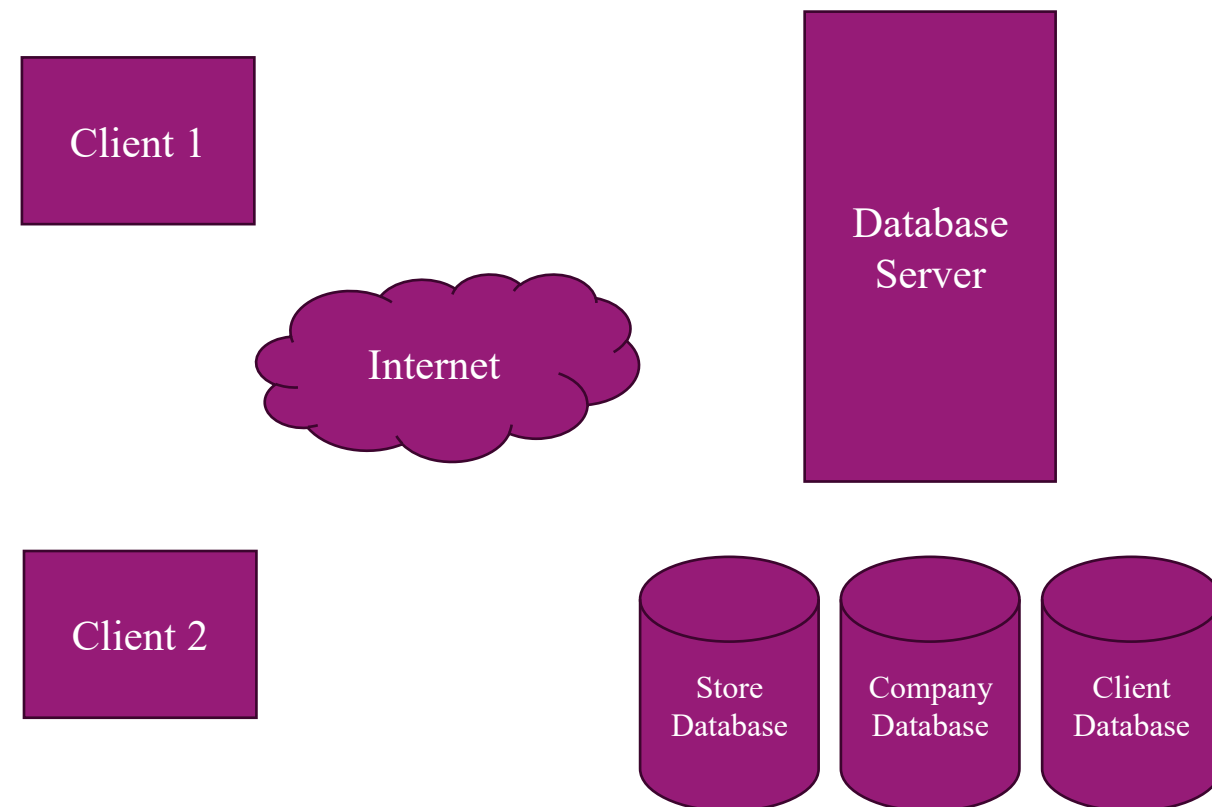
Source: <https://db-engines.com/en/ranking>

431 systems in ranking, April 2026

Rank			DBMS	Database Model	Score		
Apr 2026	Mar 2026	Apr 2025			Apr 2026	Mar 2026	Apr 2025
1.	1.	1.	Oracle	Relational, Multi-model ⓘ	1157.93	-24.53	-73.12
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6.	6.	6.	Snowflake	Relational	217.38	+6.13	+49.28
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10.	↑ 11.	↑ 11.	Apache Cassandra	Wide column, Multi-model ⓘ	102.62	+0.74	-5.59
11.	↓ 10.	↓ 8.	Elasticsearch	Multi-model ⓘ	99.51	-4.07	-28.57
12.	12.	↓ 10.	SQLite	Relational	95.90	-0.08	-18.19
13.	13.	↑ 14.	MariaDB ⓘ	Relational, Multi-model ⓘ	86.41	-0.59	-7.93
14.	14.	↑ 16.	Microsoft Azure SQL Database	Relational, Multi-model ⓘ	75.96	+2.03	+0.59
15.	15.	↑ 18.	Apache Hive	Relational	74.13	+1.18	+1.48
16.	16.	↑ 17.	Splunk	Search engine	73.08	+0.87	-0.31
17.	17.	↓ 13.	Microsoft Access	Relational	64.92	-2.65	-30.45
18.	18.	↓ 15.	Amazon DynamoDB	Multi-model ⓘ	62.67	-2.75	-13.82
19.	19.	19.	Google BigQuery	Relational	52.36	-3.46	-8.37
20.	20.	20.	Neo4j	Graph	46.88	-0.44	-0.71
21.	↑ 22.	↑ 23.	SAP HANA	Relational, Multi-model ⓘ	33.23	+1.14	-0.17
22.	↓ 21.	↑ 24.	Apache Solr	Search engine, Multi-model ⓘ	33.17	+0.38	+0.77
23.	23.	↓ 22.	Teradata	Relational, Multi-model ⓘ	31.11	+0.03	-2.97
24.	24.	↓ 21.	FileMaker	Relational	28.81	-0.50	-7.67
25.	25.	25.	SAP Adaptive Server	Relational, Multi-model ⓘ	25.31	-0.22	-3.48
26.	↑ 28.	↑ 31.	ClickHouse	Relational, Multi-model ⓘ	22.87	+0.86	+4.14
27.	↓ 26.	↓ 26.	Microsoft Azure Cosmos DB	Multi-model ⓘ	22.43	-0.31	-0.12

3. Database Architecture

- Suppose we have a "Company" database on a remote Server on the Internet and we are "Client 1"
- What are the steps to follow to extract data from a database?
- What is the important information needed to get this data?
- What to do to add security to this architecture?



4. Secure Database Architecture

- **MSSQL Server**
 - Port 1433/TCP

The default port **should** be changed because the port 1433 is the "classic" port for MSSQL Server

Source: <https://learn.microsoft.com/en-us/sql/database-engine/configure-windows/configure-a-server-to-listen-on-a-specific-tcp-port?view=sql-server-ver16>,
<https://techcommunity.microsoft.com/t5/sql-server-blog/sql-server-ports/ba-p/384096>

- **MySQL (and MariaDB)**
 - Port 3306/TCP

The default port **should** be changed because the port 3306 is the "classic" port for MySQL (and MariaDB)

Source: <https://dev.mysql.com/blog-archive/mysql-guide-to-ports/>,
<https://kinsta.com/knowledgebase/mysql-port/>

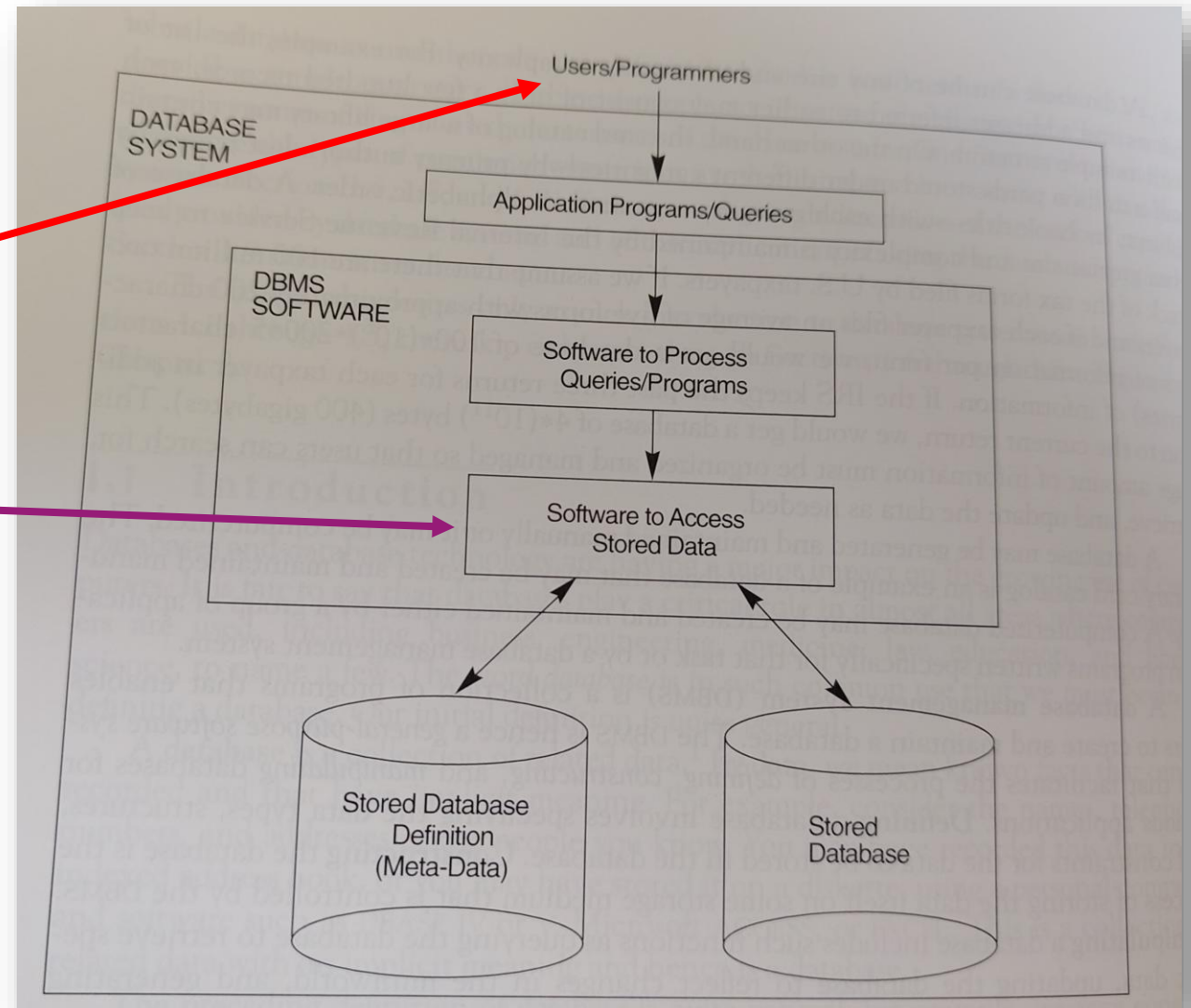
5. Keys to security

- Authentication: → Who are you?
- Authorization: → What can you do?
- Encryption: → Storing Secret Data
- Connection Security: → Restricting and Securing
- Auditing: → Recording Access
- SQL Injection: → Malicious code is inserted into SQL queries string and run on the server

5. Keys to security

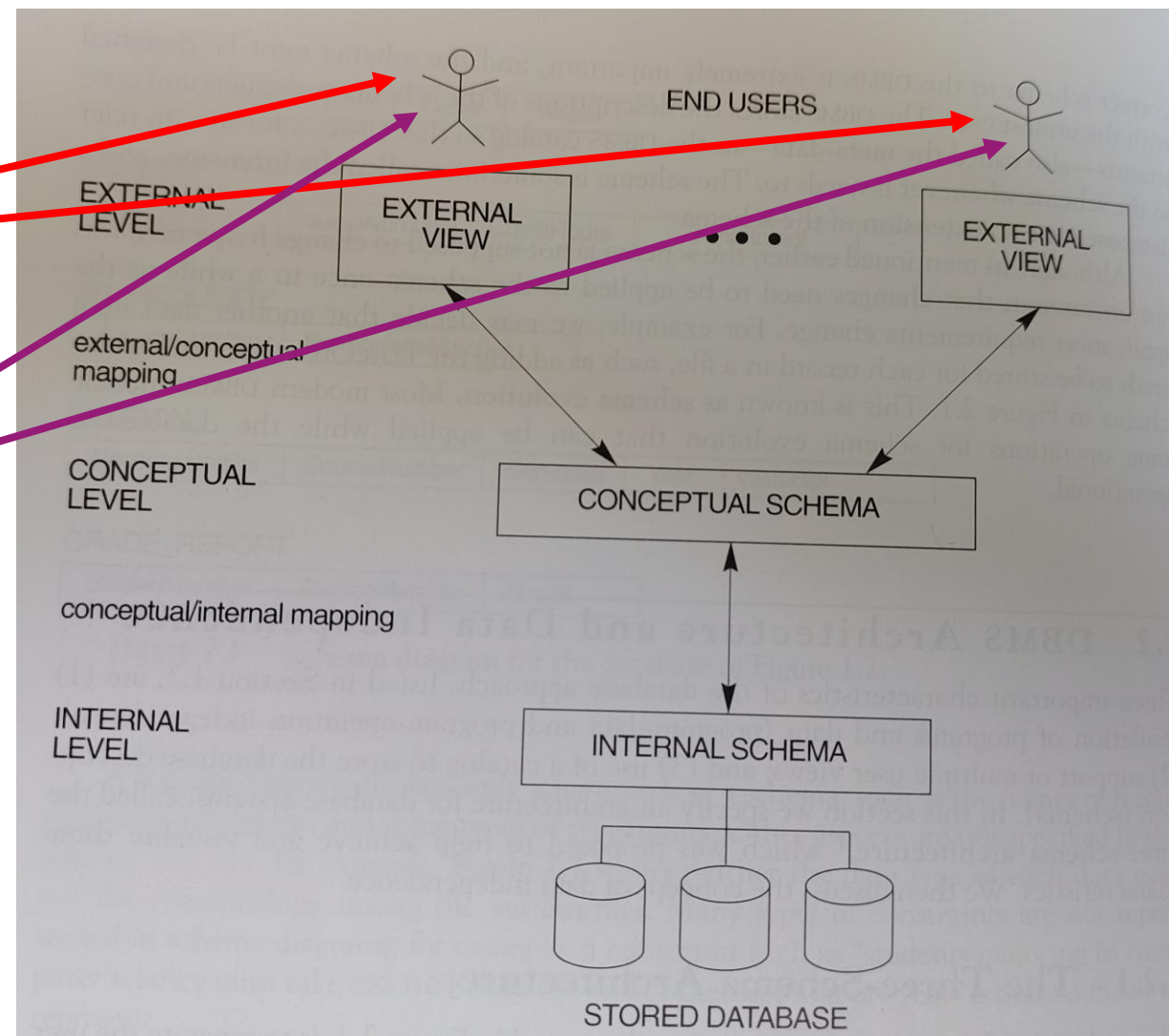
- **Authentication:**
Who are you?

- **Authorization:**
What can you do?



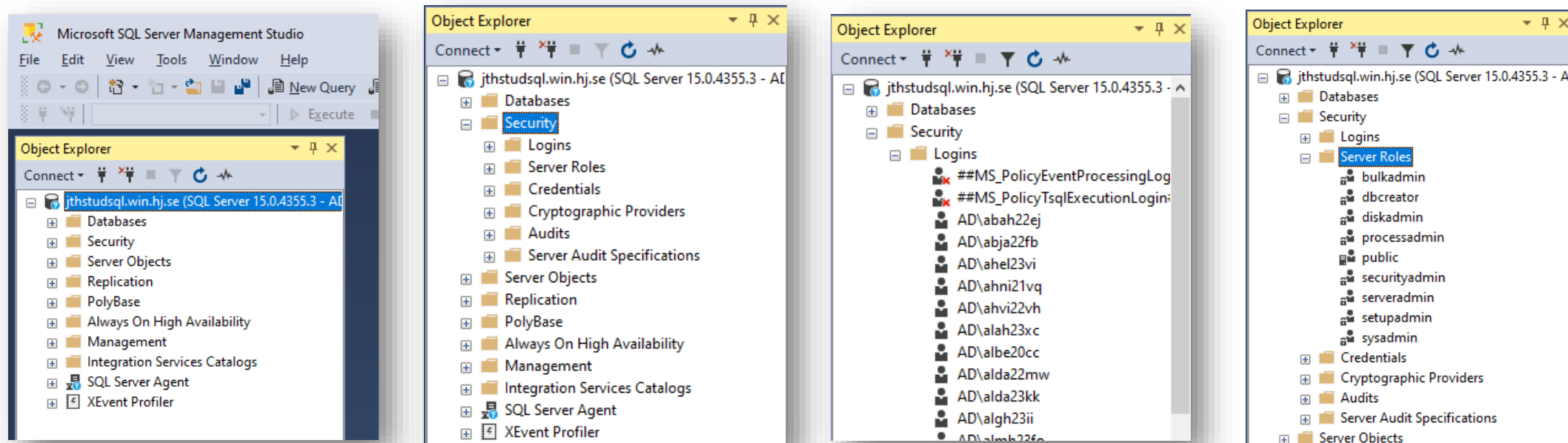
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- **Authentication:**
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- **Authorization:**
What can you do?



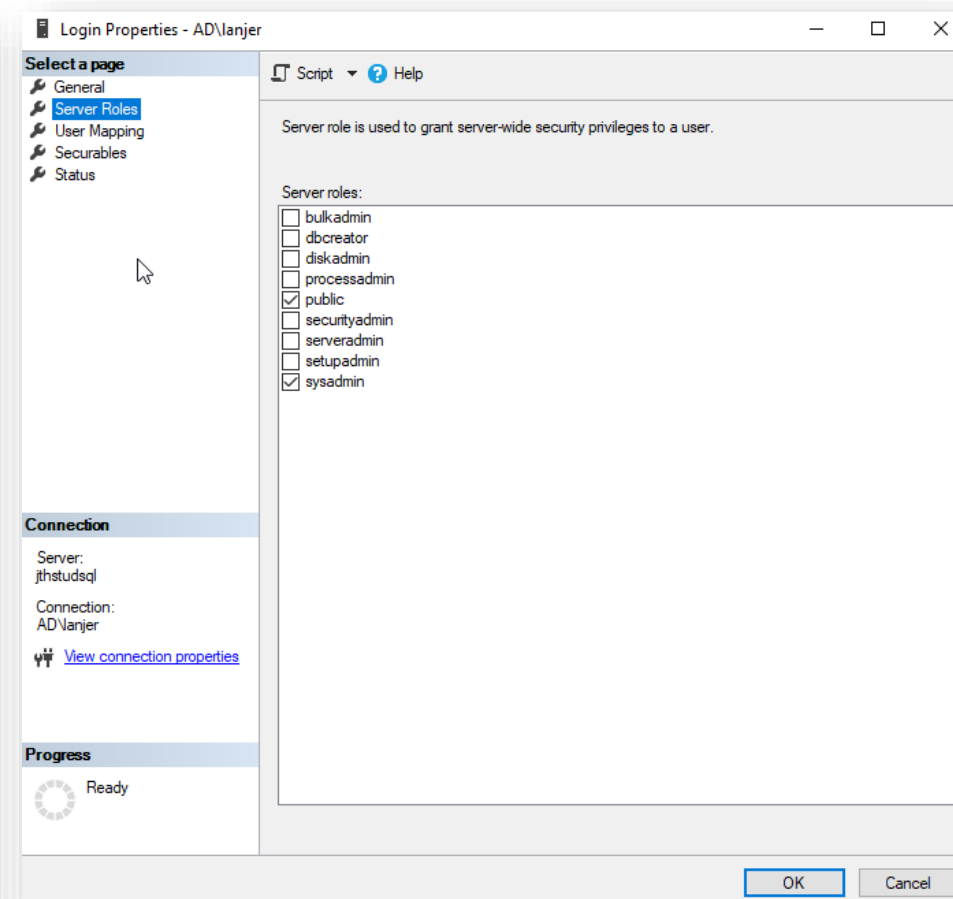
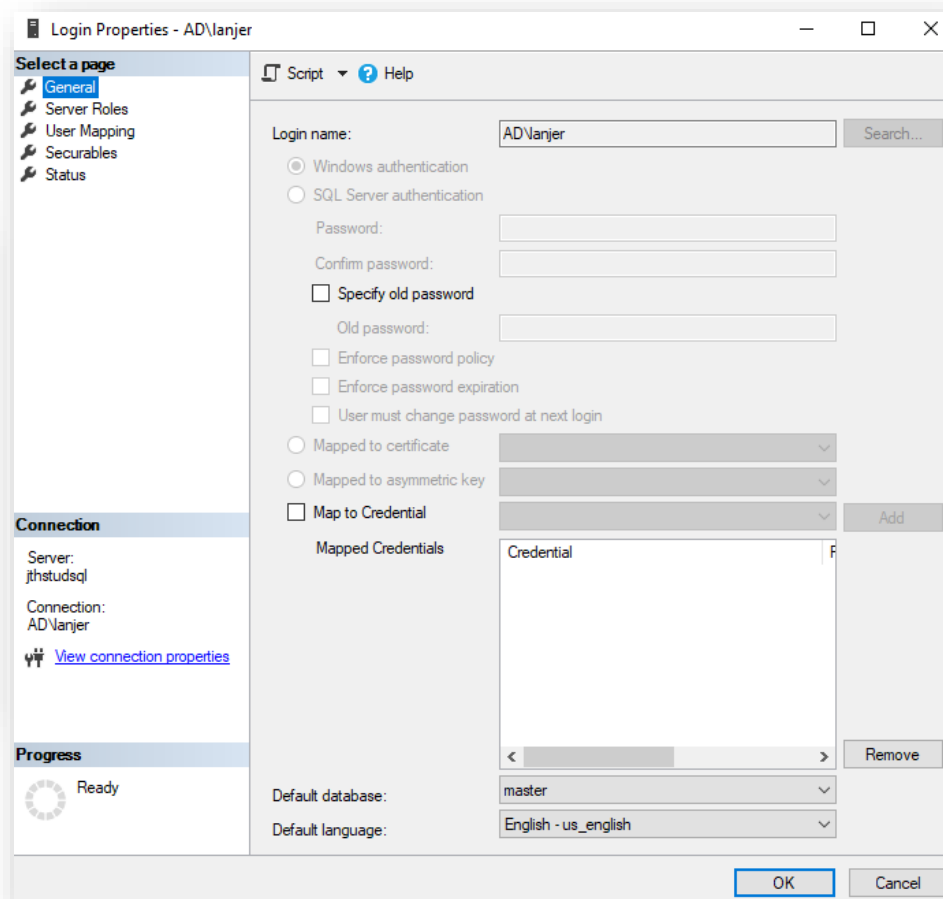
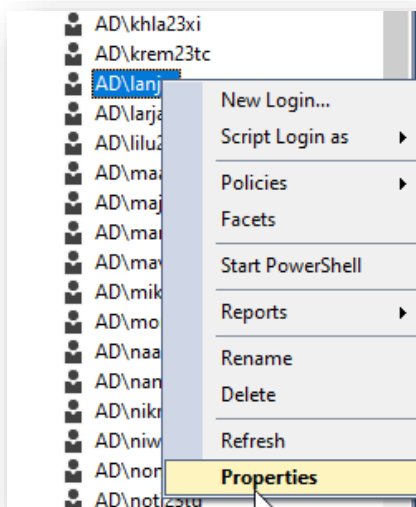
6. Permissions in MS SQL Server

- The Permissions in MS SQL Server are stored in the Database itself
- Authentication can be made from Windows Authentication (integration to the Microsoft Active Directory Environment) or using a login and password

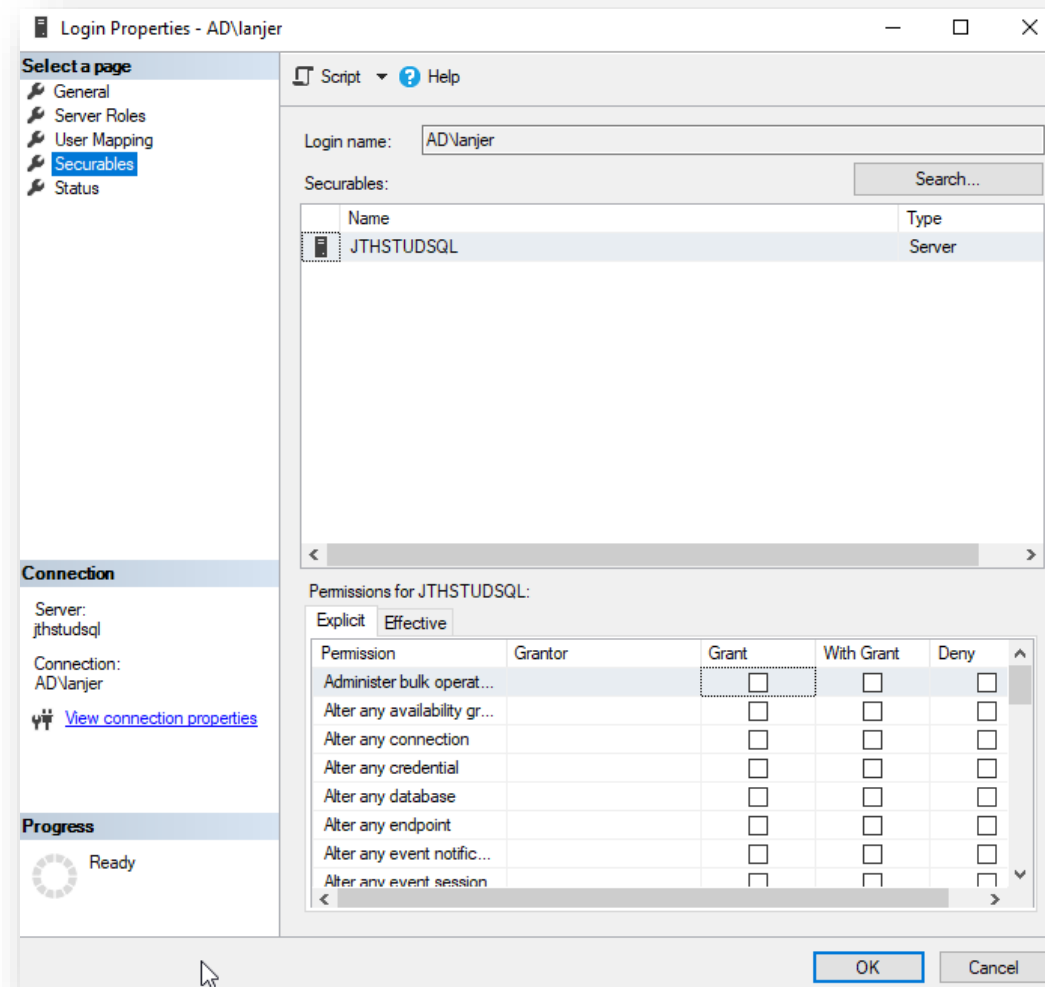
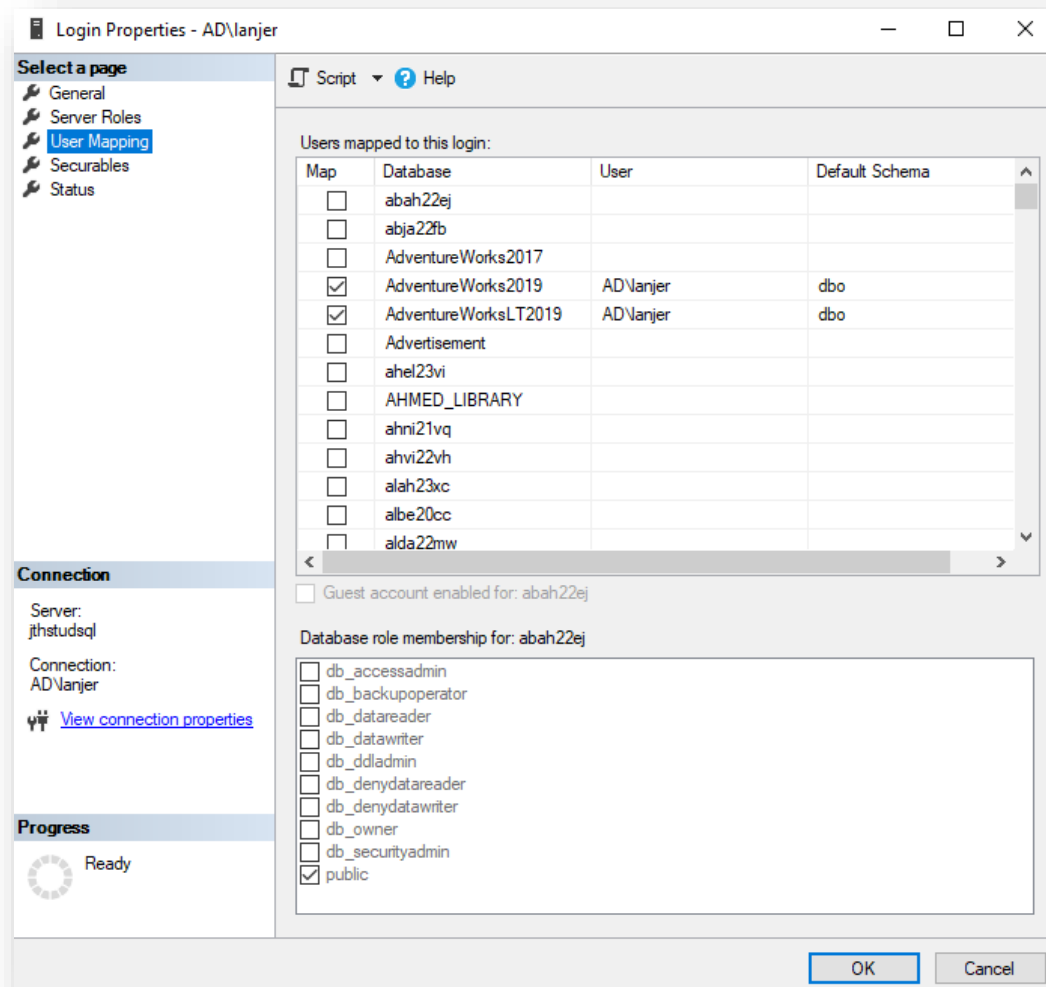


Source: <https://learn.microsoft.com/en-us/sql/relational-databases/security/security-center-for-sql-server-database-engine-and-azure-sql-database?view=sql-server-ver16>,
<https://www.guru99.com/sql-server-create-user.html>

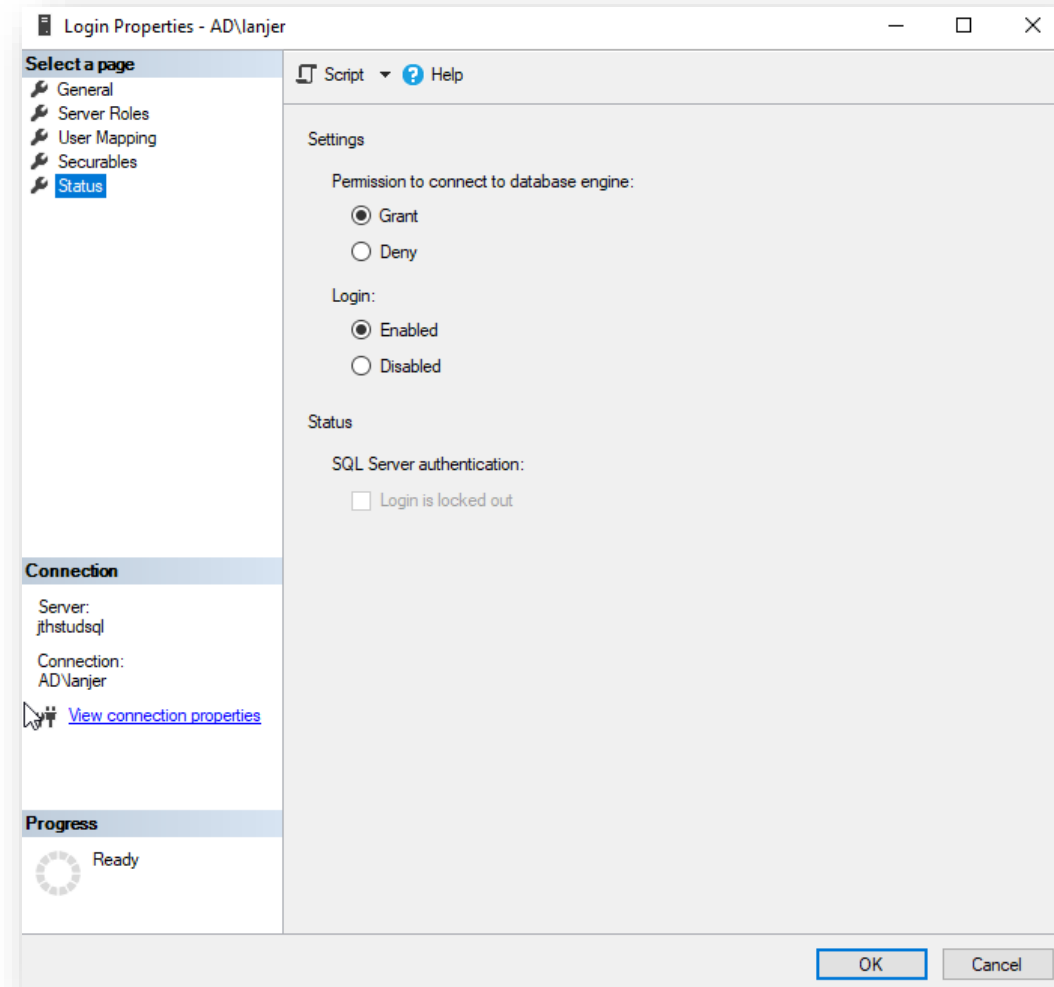
6. Permissions in MS SQL Server



6. Permissions in MS SQL Server

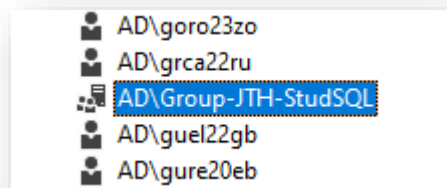


6. Permissions in MS SQL Server



6. Permissions in MS SQL Server

- If we look to an Active Directory group, we can find the permissions associated:



Login Properties - AD\Group-JTH-StudSQL

Select a page: General, Server Roles, User Mapping, Securables, Status

Script ? Help

Login name: AD\Group-JTH-StudSQL Search...

☒ Windows authentication
☐ SQL Server authentication

Password:
 Confirm password:
☐ Specify old password
 Old password:

☐ Enforce password policy
☐ Enforce password expiration
☐ User must change password at next login

☐ Mapped to certificate
☐ Mapped to asymmetric key

☐ Map to Credential Add

Mapped Credentials

Credential

Remove

Default database: master

Default language: English - us_english

OK Cancel

Connection

Server: jthstudsql

Connection: AD Vanjer

[View connection properties](#)

Progress

Ready

Login Properties - AD\Group-JTH-StudSQL

Select a page: General, Server Roles, User Mapping, Securables, Status

Script ? Help

Server role is used to grant server-wide security privileges to a user.

Server roles:

☐	bulkadmin
☐	dbcreator
☐	diskadmin
☐	processadmin
☒	public
☐	securityadmin
☐	serveradmin
☐	setupadmin
☐	sysadmin

Login Properties - AD\Group-JTH-StudSQL

Select a page: General, Server Roles, User Mapping, Securables, Status

Script ? Help

Users mapped to this login:

Map	Database	User
☐	abah22ej	
☐	abja22fb	
☒	AdventureWorks2017	AD\Group-JTH-StudSQL
☒	AdventureWorks2019	AD\Group-JTH-StudSQL
☒	AdventureWorksLT2019	AD\Group-JTH-StudSQL
☐	Advertisement	
☐	ahel23vi	

6. Permissions in MS SQL Server

- You can also create logins, passwords, users and change their rights using Transactions in SQL:
- <https://learn.microsoft.com/en-us/sql/relational-databases/security/authentication-access/determining-effective-database-engine-permissions?view=sql-server-ver16>

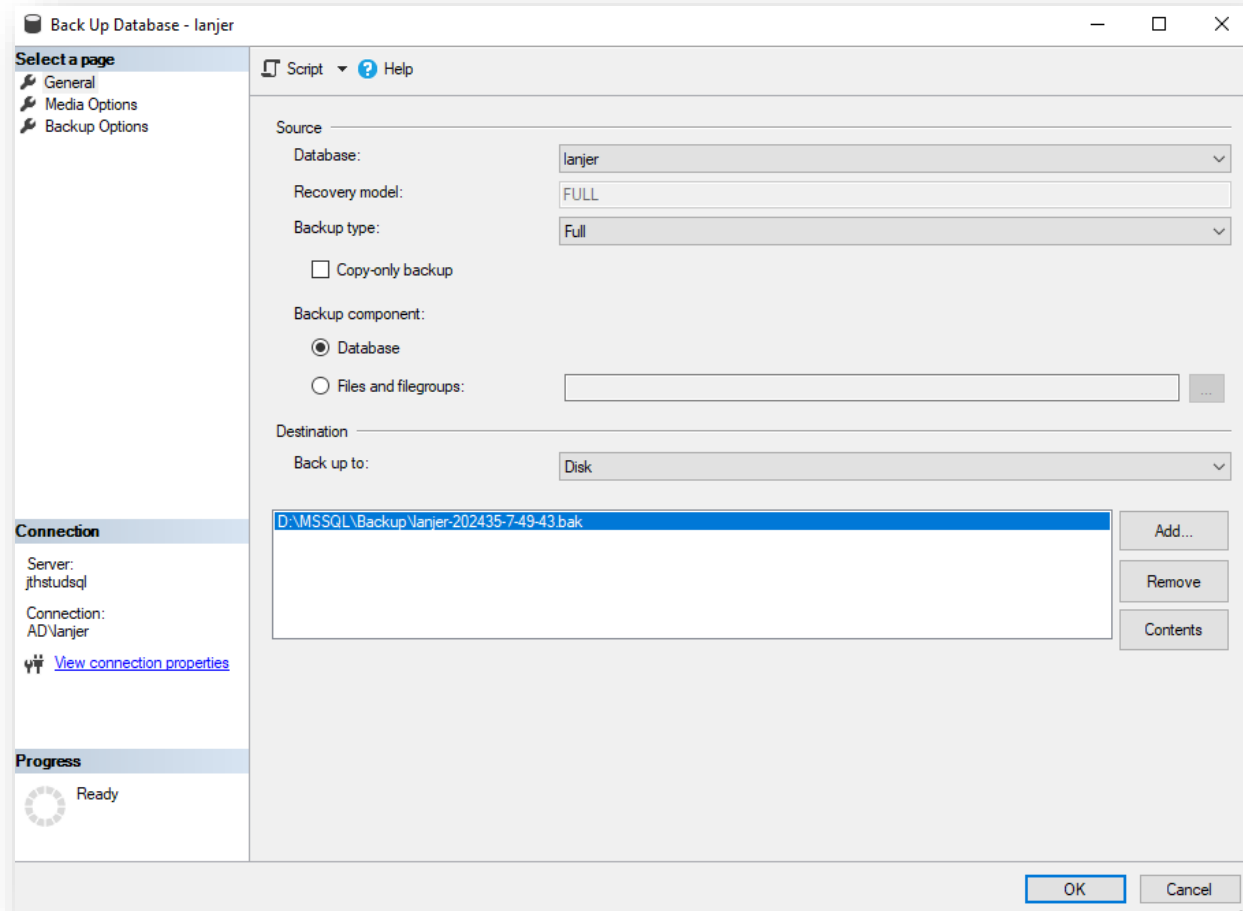
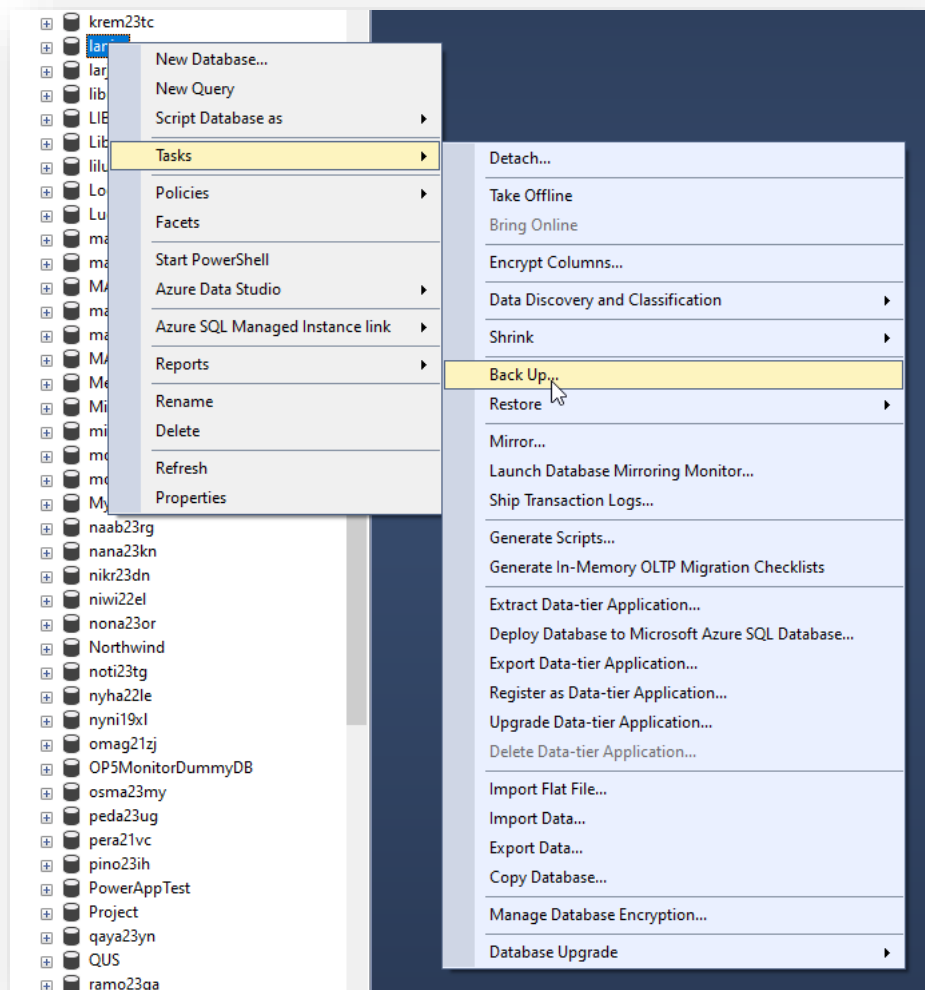
OPING UNIVERSITY
Engineering



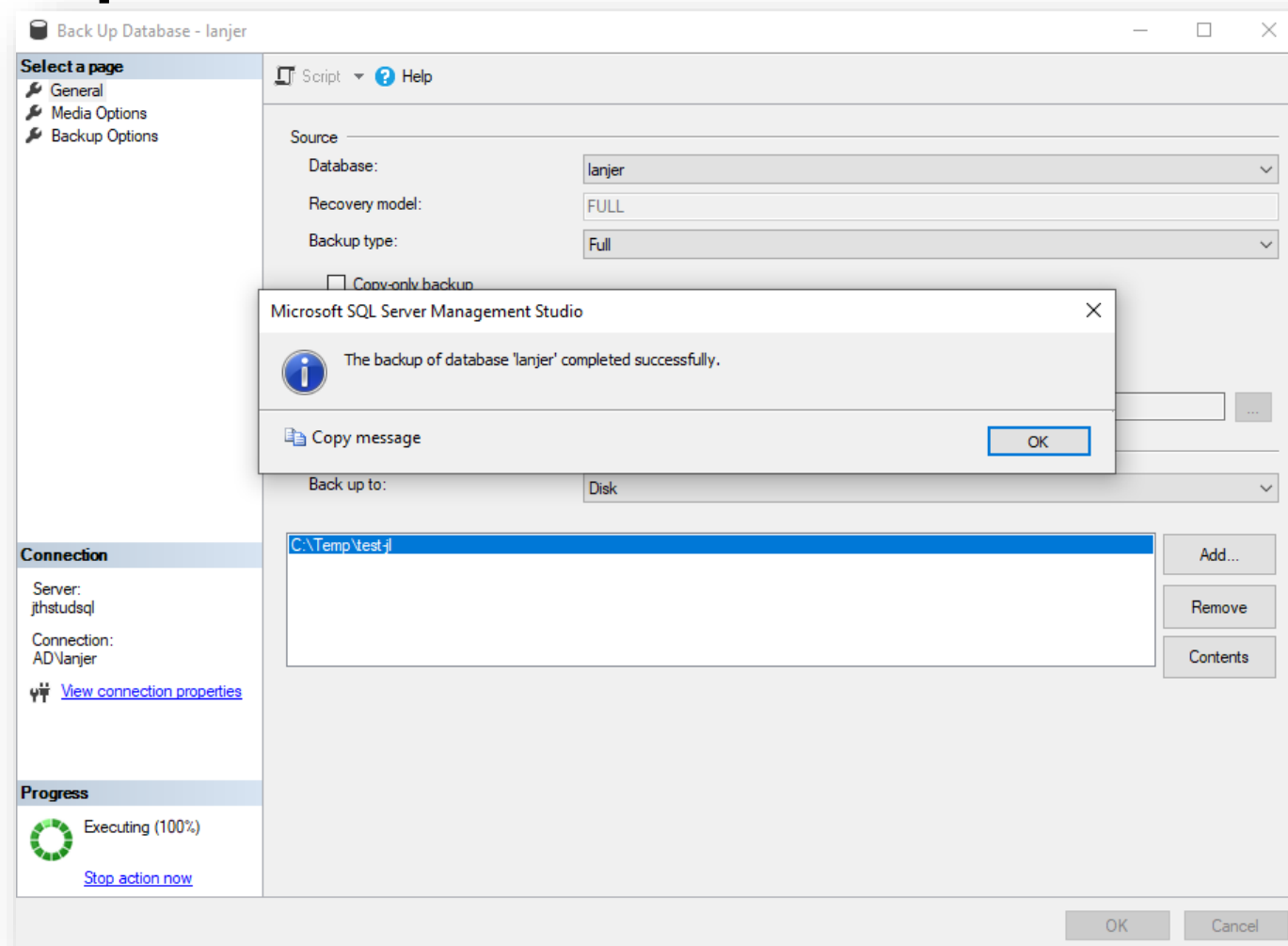
7. Backups in MS SQL Server

- Backups to ".bak" files are possible in MS SQL Server (Microsoft SQL Management Server)
- <https://learn.microsoft.com/en-us/sql/relational-databases/backup-restore/create-a-full-database-backup-sql-server?view=sql-server-ver16>

7. Backups in MS SQL Server



7. Backups in MS SQL Server



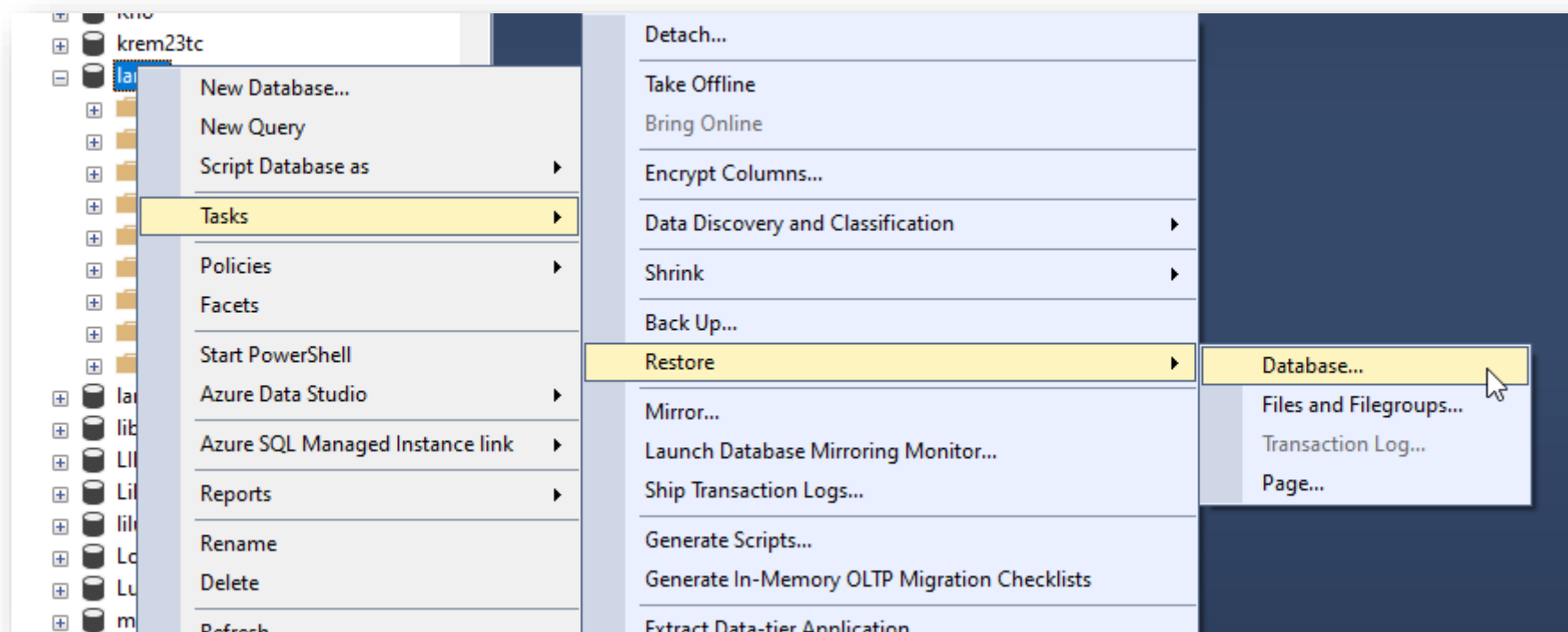
7. Backups in MS SQL Server

- Backups to ".bak" files are also possible using transaction SQL:

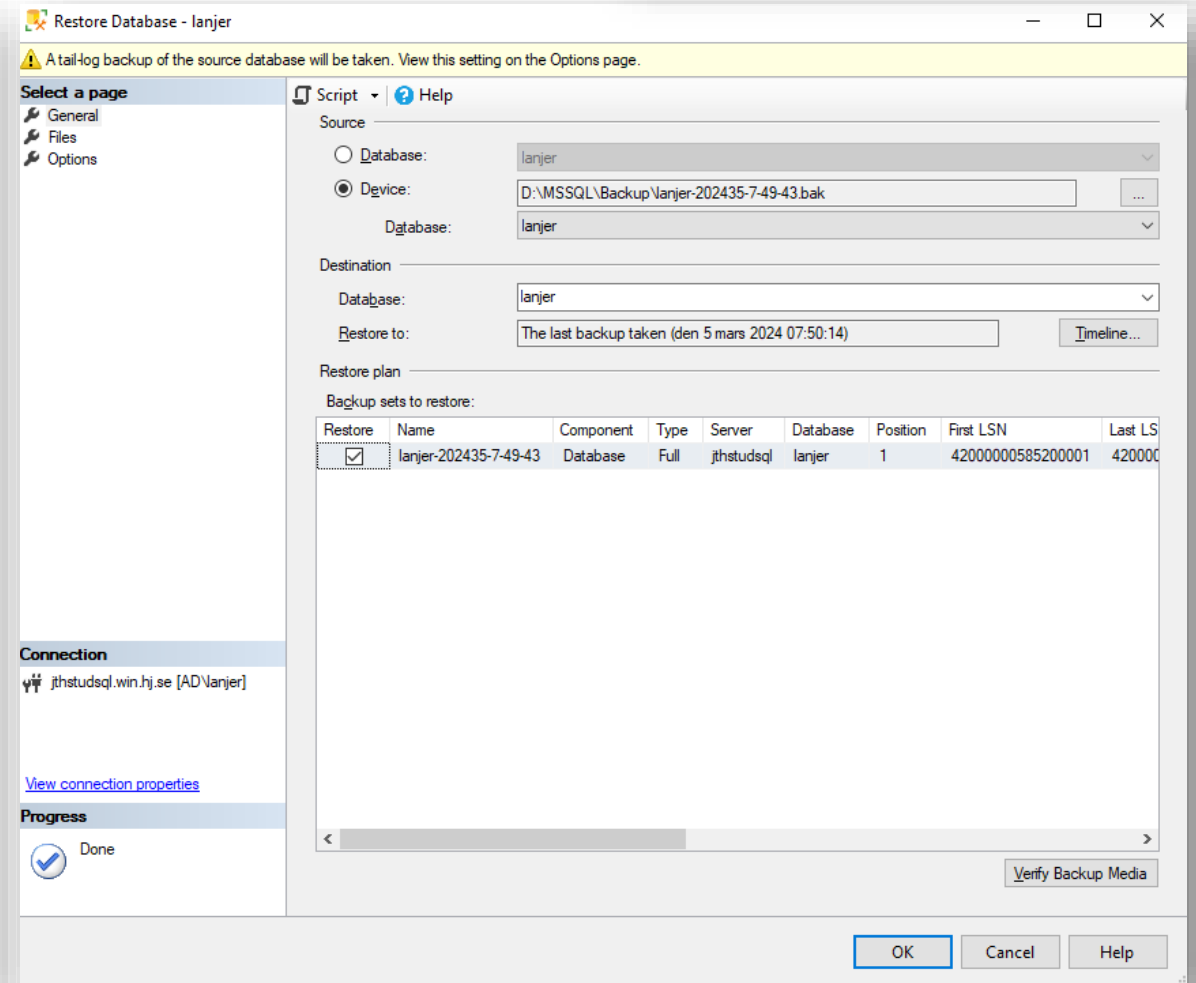
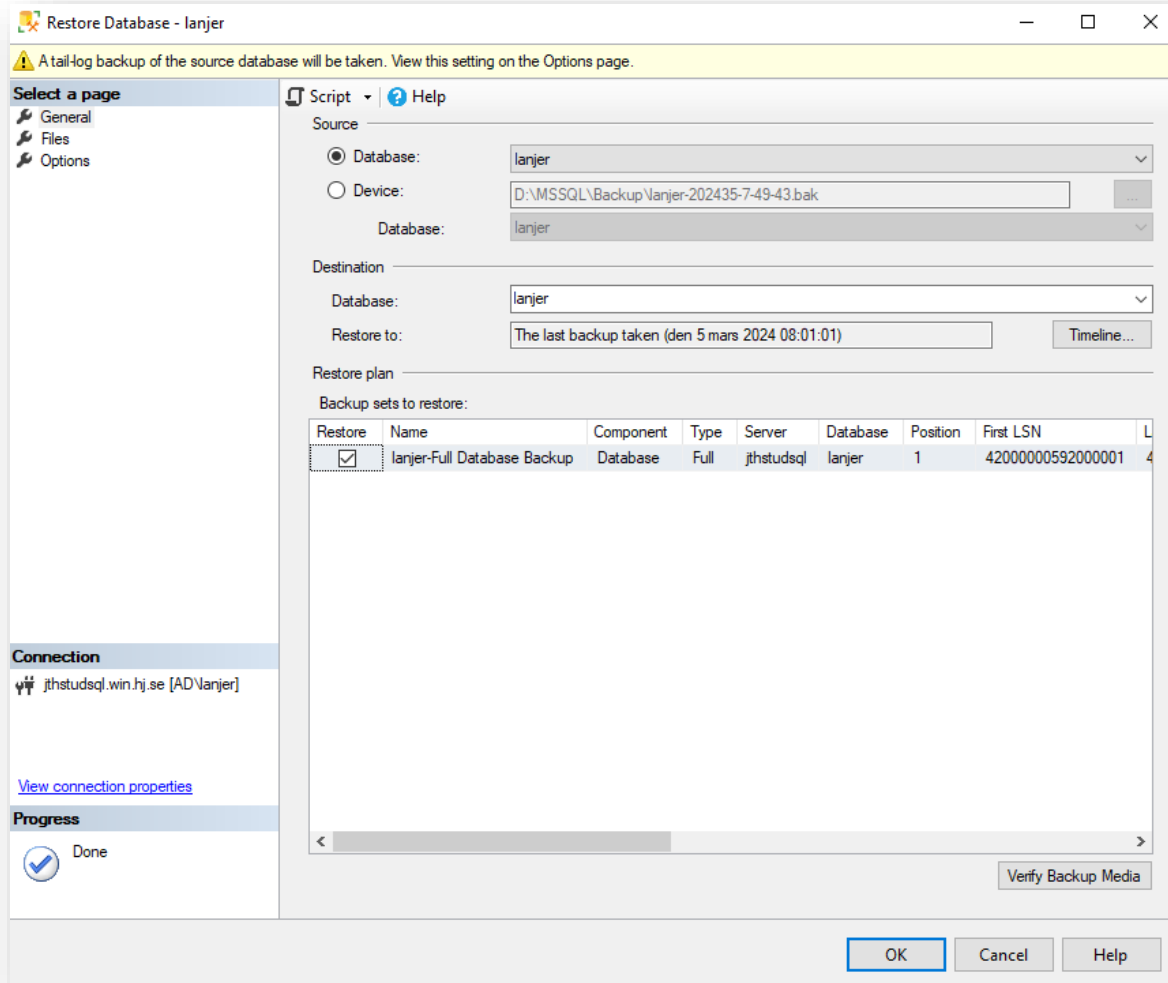
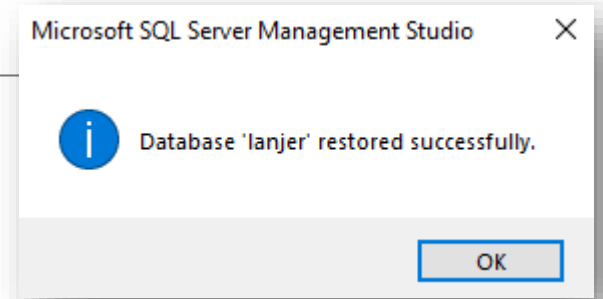
```
USE SQLTestDB;  
GO  
BACKUP DATABASE SQLTestDB  
TO DISK = 'c:\tmp\SQLTestDB.bak'  
    WITH FORMAT,  
        MEDIANAME = 'SQLServerBackups',  
        NAME = 'Full Backup of SQLTestDB';  
GO
```

7. Backups in MS SQL Server

- Backups can also be restored!



7. Backups in MS SQL Server



8. Permissions in MySQL (and MariaDB)

- MySQL is a database from Oracle (<https://en.wikipedia.org/wiki/MySQL>)
- MariaDB is the "opensource" version of MySQL before it was bought by Oracle (<https://en.wikipedia.org/wiki/MariaDB>)
- These two databases are quite similar but their development branches are different
- The permissions in MySQL (and MariaDB) Server are stored in the Database itself
- Mariadb: <https://mariadb.com/kb/en/mariadb-authorization-and-permissions-for-sql-server-users/>
- MySQL: <https://dev.mysql.com/doc/refman/8.0/en/access-control.html>

8. Permissions in MySQL (and MariaDB)

```
-- specifying plain passwords:  
CREATE USER tom@'%.example.com' IDENTIFIED BY 'plain secret';  
ALTER USER tom@'%.example.com' IDENTIFIED BY 'plain secret';  
SET PASSWORD = 'plain secret';  
  
-- specifying hashes:  
CREATE USER tom@'%.example.com' IDENTIFIED BY PASSWORD 'secret hash';  
ALTER USER tom@'%.example.com' IDENTIFIED BY PASSWORD 'secret hash';  
SET PASSWORD = PASSWORD('secret hash');
```

Permissions

Permissions can be granted to accounts. As mentioned before, the specified accounts can actually be patterns, and multiple accounts may match a pattern. For example, in this example we are creating three accounts, and we are assigning permissions to all of them:

```
CREATE USER 'tom'@'example.com';  
CREATE USER 'tom'@'123.123.123.123';  
CREATE USER 'tom'@'tomlaptop';  
GRANT USAGE ON *.* TO tom@'%';
```

8. Permissions in MySQL (and MariaDB)

The following permission levels exist in MariaDB:

- Global privileges;
- Database privileges;
- Table privileges;
- Column privileges;
- Function and procedure privileges.

```
CREATE USER 'dba'@'%' IDENTIFIED BY 'strongpassword';
GRANT ALL PRIVILEGES ON *.* TO 'dba'@'%' ;
```

Database Name

Table Name

```
SELECT USER(), CURRENT_USER();
```

```
+-----+-----+
| USER() | CURRENT_USER() |
+-----+-----+
| alice@localhost | alice@localhost |
+-----+-----+
```

```
SHOW GRANTS;
```

```
+-----+
+-----+
| Grants for alice@localhost
|
+-----+
+-----+
| GRANT ALL PRIVILEGES ON *.* TO 'alice'@'localhost' IDENTIFIED BY PASSWORD
'|*2470C0C06DEE42FD1618BB99005ADCA2EC9D1E19' |
+-----+
+-----+
```

8. Permissions in MySQL (and MariaDB)

```
sudo mysql -u root
```

```
sudo mysql -u root -p
```

```
CREATE DATABASE 'yourDB';
```

```
MariaDB [(none)]> SHOW DATABASES;
+-----+
| Database |
+-----+
| example |
| information_schema |
| mysql |
| performance_schema |
| yourDB |
+-----+
5 rows in set (0.000 sec)
```

```
MariaDB [(none)]> CREATE USER 'user1'@localhost IDENTIFIED BY 'password1'; 1
Query OK, 0 rows affected (0.000 sec)
MariaDB [(none)]> SELECT User FROM mysql.user; 2
+-----+
| User |
+-----+
| newuser |
| user1 |
| root |
| root |
+-----+
8 rows in set (0.000 sec)
MariaDB [(none)]>
```

```
GRANT ALL PRIVILEGES ON *.* TO 'user1'@localhost IDENTIFIED BY 'password1';
```

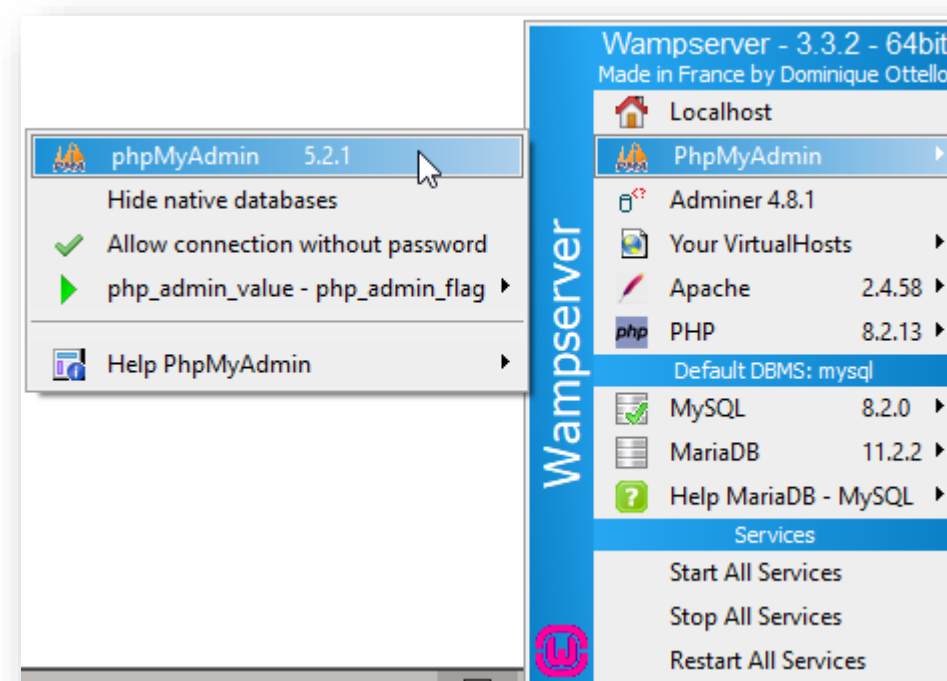
```
MariaDB [(none)]> SHOW GRANTS FOR 'user1'@localhost;
+-----+
| Grants for user1@localhost |
+-----+
| GRANT ALL PRIVILEGES ON *.* TO 'user1'@'localhost' IDENTIFIED BY PASSWORD '*668425423DB5193AF921380129F465A6425216D0' |
+-----+
1 row in set (0.000 sec)
MariaDB [(none)]>
```

```
GRANT ALL PRIVILEGES ON 'yourDB'.* TO 'user1'@localhost;
```

OR

9. Backups in MySQL (and MariaDB)

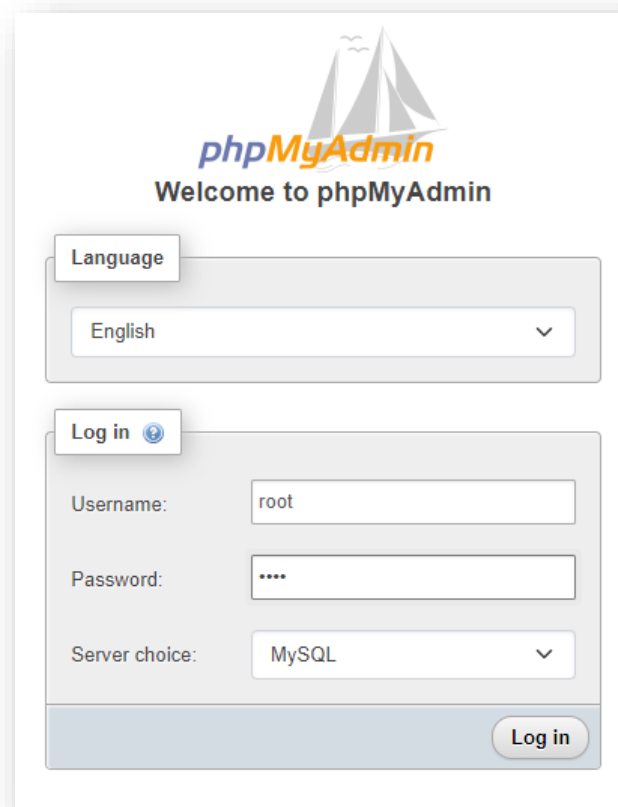
- Backups are made by extracting information and data from the tables using a tool such as PHPMyAdmin
- You can install a full webserver development package with WAMP Server



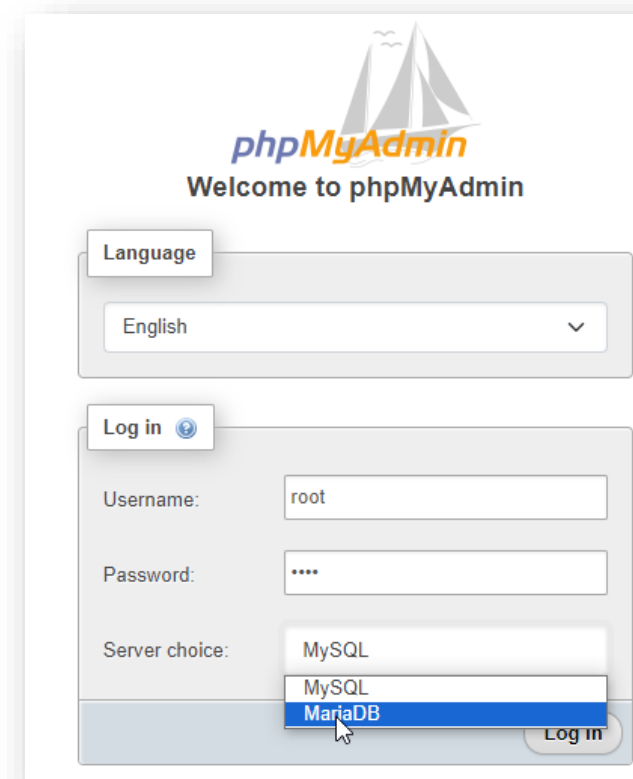
9. Backups in MySQL (and MariaDB)

- Backups are made by extracting information and data from the tables using a tool such as PHPMyAdmin

`http://localhost/phpmyadmin/`

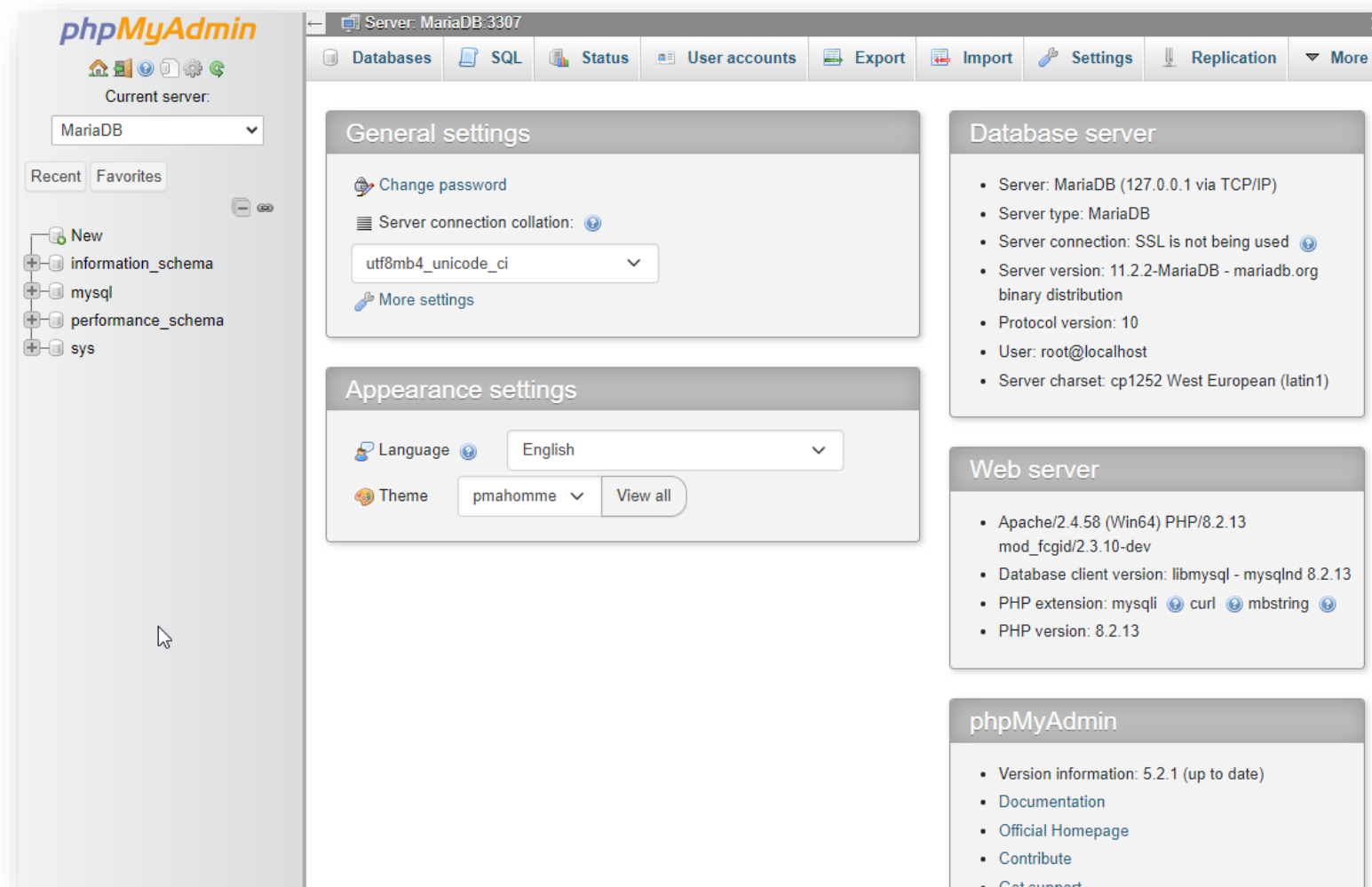


The screenshot shows the phpMyAdmin login interface. At the top is the phpMyAdmin logo and the text "Welcome to phpMyAdmin". Below this is a "Language" dropdown menu set to "English". Underneath is a "Log in" button with a help icon. The login form contains three fields: "Username:" with the value "root", "Password:" with masked characters "****", and "Server choice:" with a dropdown menu set to "MySQL". A "Log in" button is located at the bottom right of the form.



This screenshot is similar to the previous one, but the "Server choice:" dropdown menu is open, showing two options: "MySQL" and "MariaDB". The "MariaDB" option is highlighted with a blue background. The "Log in" button is visible at the bottom right.

9. Backups in MySQL (and MariaDB)



9. Backups in MySQL (and MariaDB)

phpMyAdmin

Current server: MariaDB

Recent Favorites

New

- information_schema
- mysql
- performance_schema
- sys

Server: MariaDB:3307

Databases SQL Status User accounts Export Import

Databases

Create database

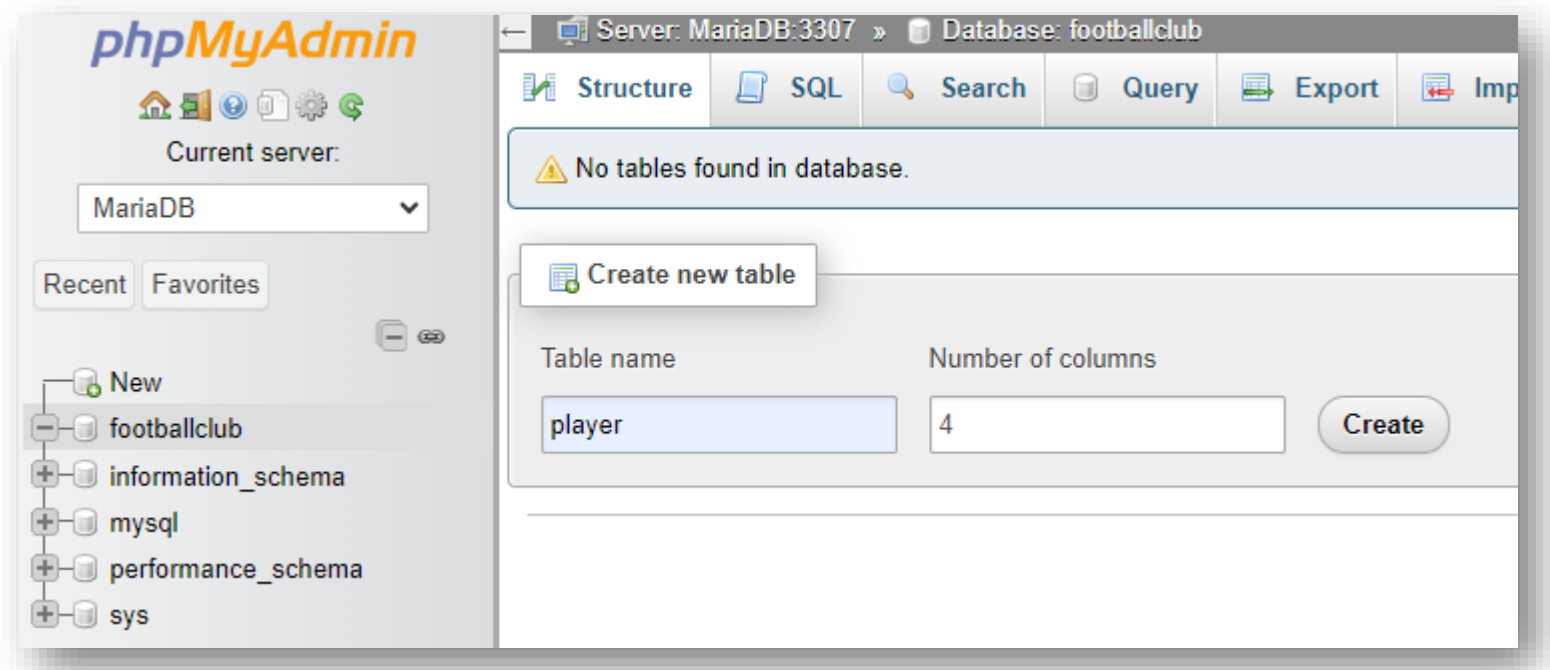
footballclub utf8mb3_unicode_ci Create

☐ Check all

Database	Collation	Action
☐ information_schema	utf8mb3_general_ci	Check privileges
☐ mysql	latin1_swedish_ci	Check privileges
☐ performance_schema	utf8mb3_general_ci	Check privileges
☐ sys	utf8mb3_general_ci	Check privileges

Total: 4

9. Backups in MySQL (and MariaDB)



9. Backups in MySQL (and MariaDB)

phpMyAdmin

Current server: MariaDB

Recent Favorites

New

- footballclub
- information_schema
- mysql
- performance_schema
- sys

Server: MariaDB:3307 » Database: footballclub

Structure SQL Search Query Export Import Operations Privileges Routines Events Triggers Designer

Table name: player Add 1 column(s) Go

Name	Type	Length/Values	Default	Collation	Attributes	Null	Index	Comments
p_id	INT		None			☐	PRIMARY	☒
p_fName	VARCHAR	30	None			☐	---	☐
p_lName	VARCHAR	50	None			☐	---	☐
pos_id_	INT		None			☐	---	☐

Table comments: Collation: Storage Engine: MyISAM

PARTITION definition:

Partition by: (Expression or column list)

Partitions:

Preview SQL Save

9. Backups in MySQL (and MariaDB)

phpMyAdmin

Current server: MariaDB

Recent Favorites

- New
- footballclub
 - New
 - player
- information_schema
- mysql
- performance_schema
- sys

Server: MariaDB:3307 » Database: footballclub » Table: player

Browse Structure SQL Search Insert Export Import Privileges Operations Triggers

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 p_id	int(11)			No	None		AUTO_INCREMENT	Change Drop More
☐	2 p_fName	varchar(30)	utf8mb3_unicode_ci		No	None			Change Drop More
☐	3 p_lName	varchar(50)	utf8mb3_unicode_ci		No	None			Change Drop More
☐	4 pos_id_	int(11)			No	None			Change Drop More

☐ Check all With selected: Browse Change Drop Primary Unique Index Spatial Fulltext

Print Propose table structure Move columns Normalize

Add 1 column(s) after pos_id_ Go

Indexes

Action	Keyname	Type	Unique	Packed	Column	Cardinality	Collation	Null	Comment
Edit Rename Drop	PRIMARY	BTREE	Yes	No	p_id	0	A	No	

Create an index on 1 columns Go

9. Backups in MySQL (and MariaDB)

phpMyAdmin

Current server: MariaDB

Recent Favorites

- New
- footballclub
 - New
 - player
- information_schema
- mysql
- performance_schema
- sys

Server: MariaDB:3307 » Database: footballclub » Table: player

Browse Structure SQL Search Insert Export Import Privileges Operations

Column	Type	Function	Null	Value
p_id	int(11)			1
p_fName	varchar(30)			Jérôme
p_lName	varchar(50)			Landré
pos_id_	int(11)			1

Go

☐ Ignore

Column	Type	Function	Null	Value
p_id	int(11)			2
p_fName	varchar(30)			Linus
p_lName	varchar(50)			Rudbeck
pos_id_	int(11)			2

Go

9. Backups in MySQL (and MariaDB)

The screenshot displays the phpMyAdmin web interface. The top navigation bar shows the current server as 'MariaDB' and the selected database as 'footballclub'. The 'Table: player' is selected, and the 'SQL' tab is active. A green message bar at the top indicates '2 rows inserted.' Below this, the executed SQL query is shown: `INSERT INTO `player` (`p_id`, `p_fName`, `p_lName`, `pos_id`) VALUES ('1', 'Jérôme', 'Landré', '1'), ('2', 'Linus', 'Rudbeck', '2');`. The main query editor contains the same query. On the right, a table structure preview for 'player' is visible, showing columns: p_id, p_fName, p_lName, and pos_id. The bottom of the interface includes buttons for query types (SELECT *, SELECT, INSERT, UPDATE, DELETE), a 'Go' button, and checkboxes for 'Bind parameters', 'Show this query here again', 'Retain query box', 'Rollback when finished', and 'Enable foreign key checks'.

phpMyAdmin

Current server: MariaDB

Recent Favorites

Server: MariaDB:3307 » Database: footballclub » Table: player

Browse Structure SQL Search Insert Export Import Privileges Operations Triggers

✓ 2 rows inserted.

`INSERT INTO `player` (`p_id`, `p_fName`, `p_lName`, `pos_id`) VALUES ('1', 'Jérôme', 'Landré', '1'), ('2', 'Linus', 'Rudbeck', '2');`

[Edit inline] [Edit] [Create PHP code]

Run SQL query/queries on table footballclub.player:

```
1 INSERT INTO `player` (`p_id`, `p_fName`, `p_lName`, `pos_id`) VALUES ('1', 'Jérôme', 'Landré', '1'), ('2', 'Linus', 'Rudbeck', '2');
```

p_id
p_fName
p_lName
pos_id

SELECT * SELECT INSERT UPDATE DELETE Clear Format Get auto-saved query

☐ Bind parameters

Delimiter ; ☐ Show this query here again ☐ Retain query box ☐ Rollback when finished ☒ Enable foreign key checks Go

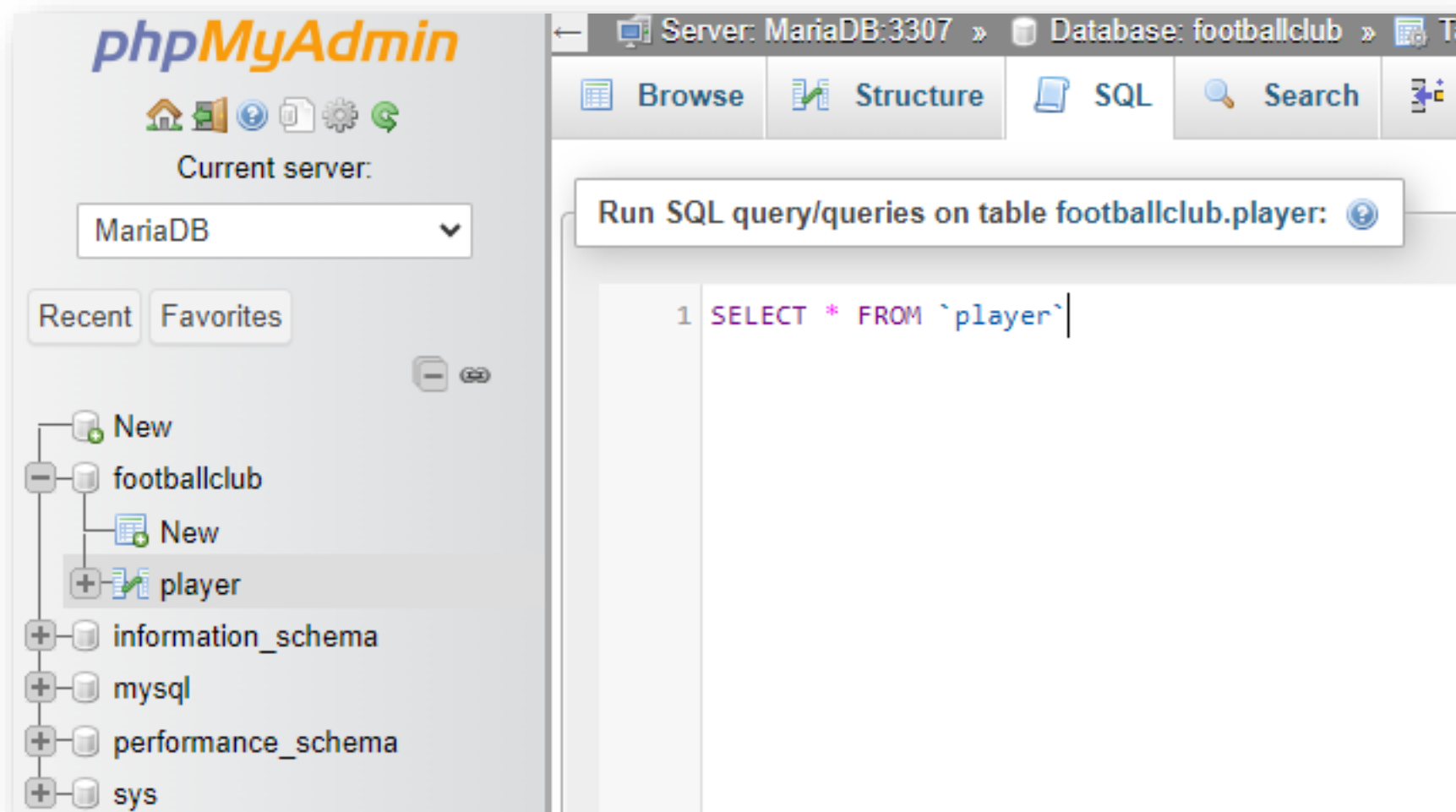
9. Backups in MySQL (and MariaDB)

The screenshot displays the phpMyAdmin web interface. The top navigation bar includes tabs for Browse, Structure, SQL, Search, Insert, Export, Import, Privileges, Operations, and Triggers. The left sidebar shows the database structure with 'footballclub' selected, containing 'player', 'information_schema', 'mysql', 'performance_schema', and 'sys'. The main panel shows the 'player' table with the following data:

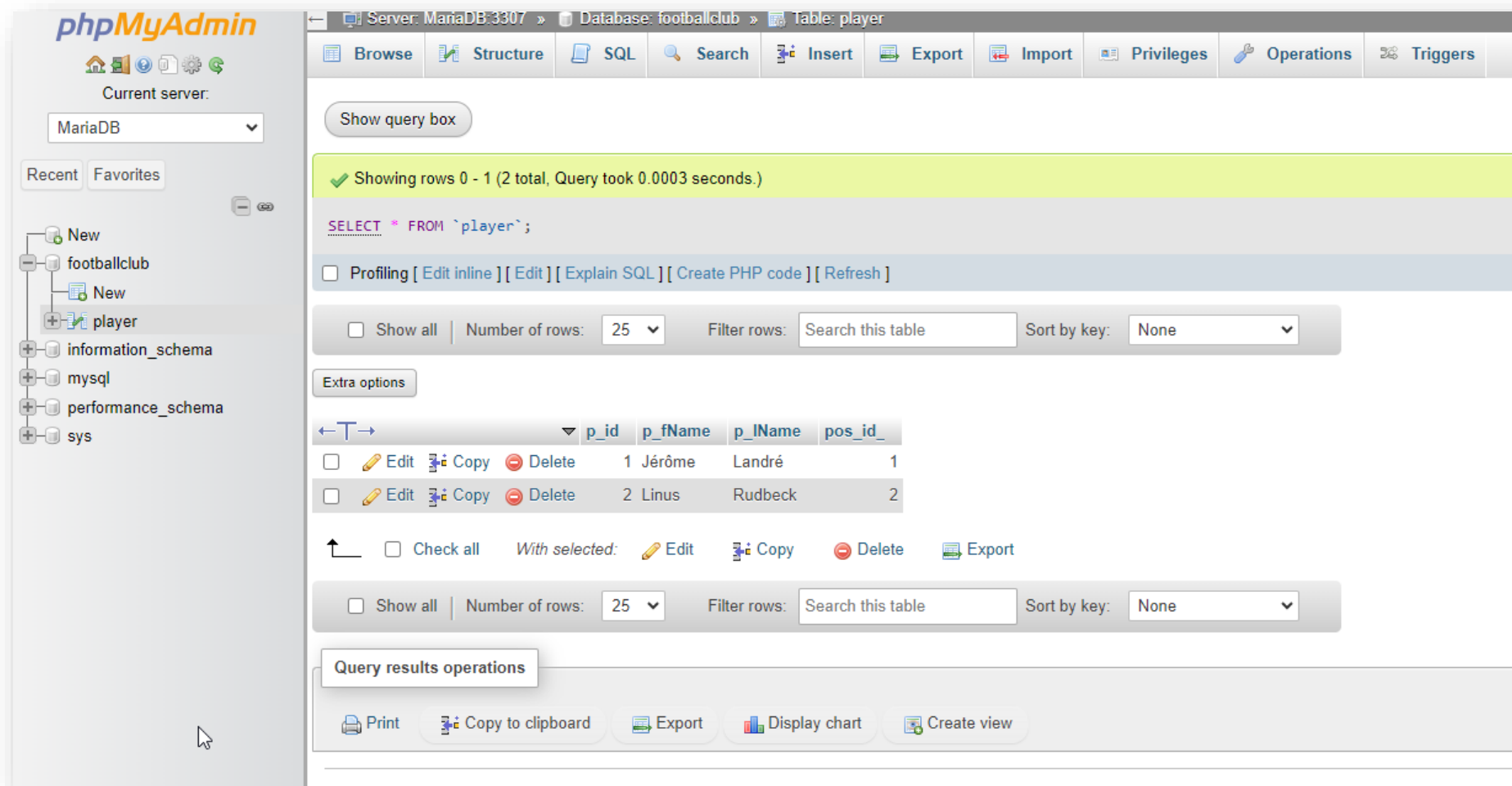
	p_id	p_fName	p_lName	pos_id
☐ Edit Copy Delete	1	Jérôme	Landré	1
☐ Edit Copy Delete	2	Linus	Rudbeck	2

Below the table, there are controls for 'Check all', 'With selected', 'Edit', 'Copy', 'Delete', and 'Export'. The bottom of the interface features a 'Query results operations' section with buttons for Print, Copy to clipboard, Export, Display chart, and Create view.

9. Backups in MySQL (and MariaDB)



9. Backups in MySQL (and MariaDB)



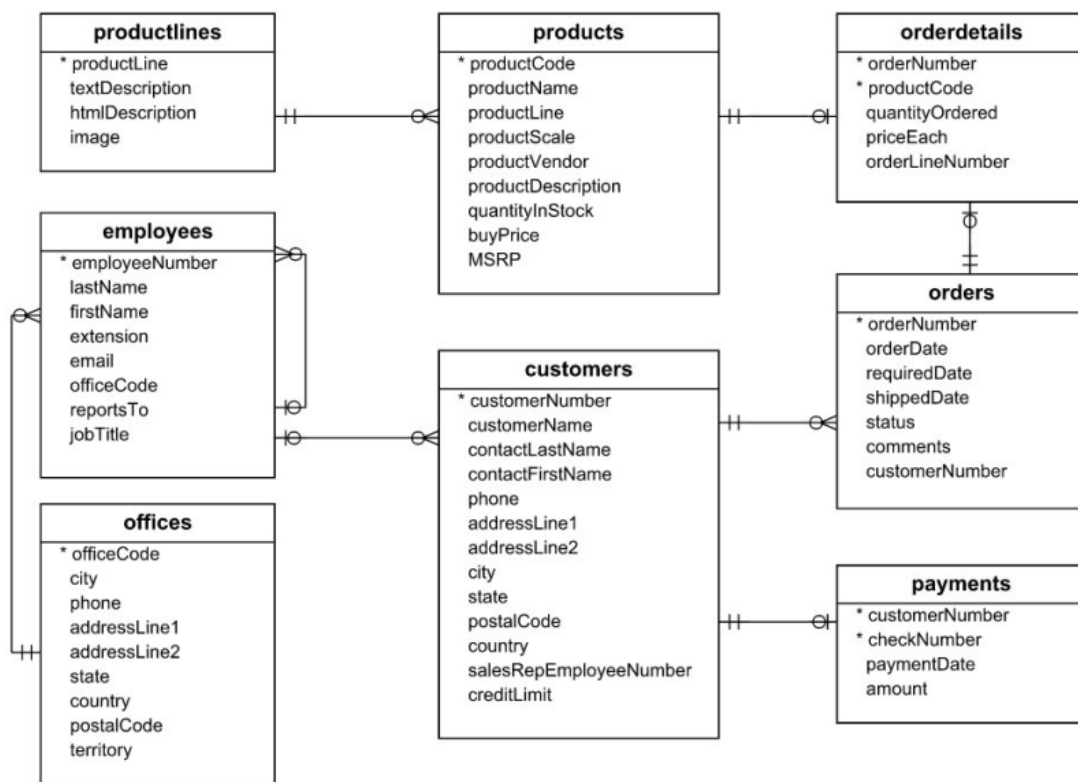
The screenshot displays the phpMyAdmin web interface. On the left sidebar, the 'Current server' is set to 'MariaDB'. The database 'footballclub' is selected, and the 'player' table is highlighted. The main panel shows the table structure and data. The SQL query 'SELECT * FROM `player`;' is executed, displaying two rows of data. The table has columns: p_id, p_fName, p_lName, and pos_id.

	p_id	p_fName	p_lName	pos_id
☐	1	Jérôme	Landré	1
☐	2	Linus	Rudbeck	2

Query results operations: Print, Copy to clipboard, Export, Display chart, Create view.

9. Backups in MySQL (and MariaDB)

MySQL Sample Database Diagram



Name	Date modified	Type	Size
mysqlsampledatabse.sql	2023-10-29 13:15	Microsoft SQL ...	192 KB

```

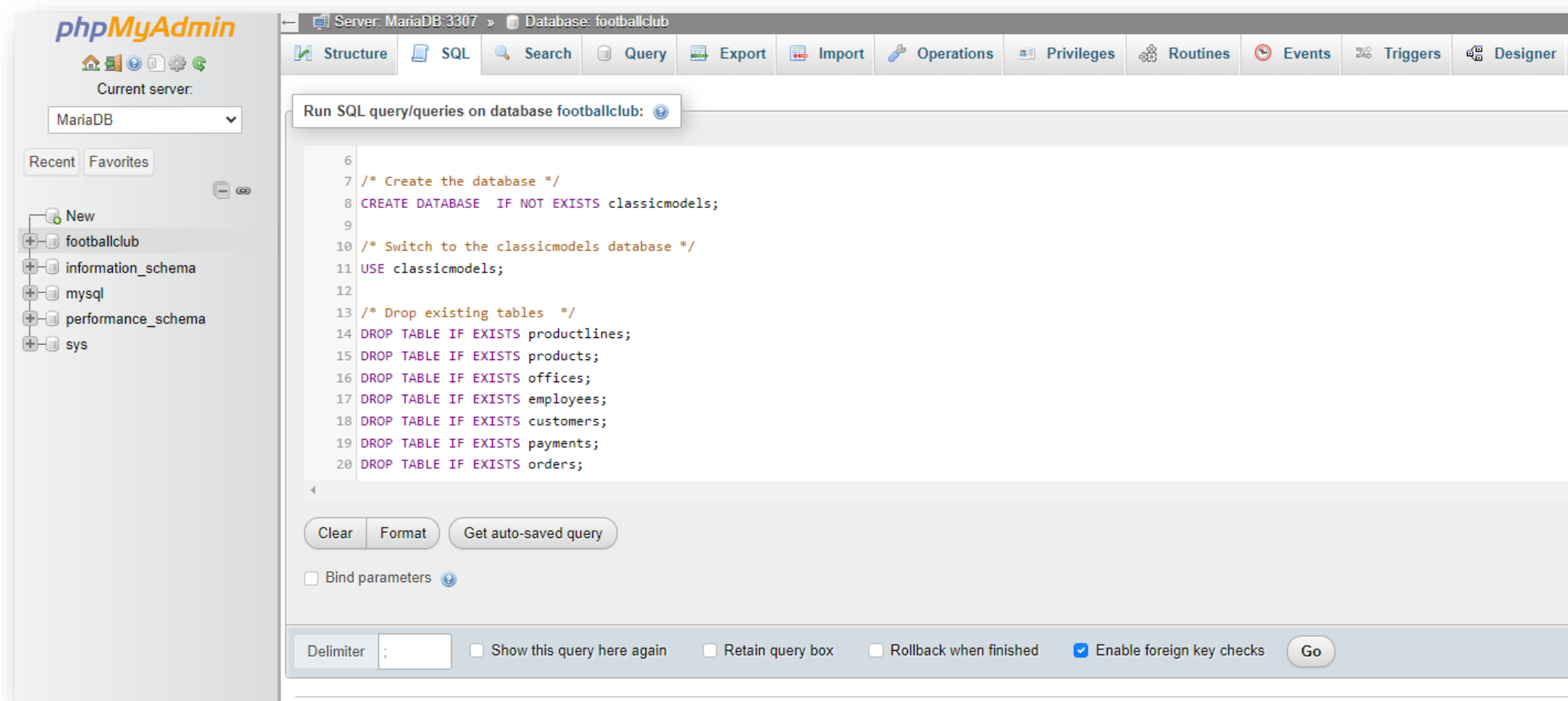
10  /* Switch to the classicmodels database */
11  USE classicmodels;
12
13  /* Drop existing tables */
14  DROP TABLE IF EXISTS productlines;
15  DROP TABLE IF EXISTS products;
16  DROP TABLE IF EXISTS offices;
17  DROP TABLE IF EXISTS employees;
18  DROP TABLE IF EXISTS customers;
19  DROP TABLE IF EXISTS payments;
20  DROP TABLE IF EXISTS orders;
21  DROP TABLE IF EXISTS orderdetails;
22
23  /* Create the tables */
24  CREATE TABLE productlines (
25      productLine varchar(50),
26      textDescription varchar(4000) DEFAULT NULL,
27      htmlDescription mediumtext,
28      image mediumblob,
29      PRIMARY KEY (productLine)
30  );
31
32  CREATE TABLE products (
33      productCode varchar(15),
34      productName varchar(70) NOT NULL,
35      productLine varchar(50) NOT NULL,
36      productScale varchar(10) NOT NULL,
37      productVendor varchar(50) NOT NULL,
  
```

Source: <https://www.mysqltutorial.org/getting-started-with-mysql/mysql-sample-database>

9. Backups in MySQL (and MariaDB)

```
59 CREATE TABLE employees (  
60     employeeNumber int,  
61     lastName varchar(50) NOT NULL,  
62     firstName varchar(50) NOT NULL,  
63     extension varchar(10) NOT NULL,  
64     email varchar(100) NOT NULL,  
65     officeCode varchar(10) NOT NULL,  
66     reportsTo int DEFAULT NULL,  
67     jobTitle varchar(50) NOT NULL,  
68     PRIMARY KEY (employeeNumber),  
69     FOREIGN KEY (reportsTo) REFERENCES employees (employeeNumber),  
70     FOREIGN KEY (officeCode) REFERENCES offices (officeCode)  
71 );  
72
```

9. Backups in MySQL (and MariaDB)



Source: <https://www.mysqltutorial.org/getting-started-with-mysql/mysql-sample-database>

Server: MariaDB:3307 » Database: classicmodels » Table: employees

Browse Structure SQL Search Insert Export Import Privileges Operations Triggers

Run SQL query/queries on database classicmodels:

```
1 SELECT * FROM employees
```

Clear Format Get auto-saved query

☐ Bind parameters

Delimiter: ; ☐ Show this query here again ☐ Retain query box ☐ Rollback when finished ☒ Enable foreign key checks Go

Hide query box

Showing rows 0 - 22 (23 total, Query took 0.0004 seconds.)

```
SELECT * FROM employees;
```

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

	employeeNumber	lastName	firstName	extension	email	officeCode	reportsTo	jobTitle
☐ Edit Copy Delete	1002	Murphy	Diane	x5800	dmurphy@classicmodelcars.com	1	NULL	President
☐ Edit Copy Delete	1056	Patterson	Mary	x4611	mpatterso@classicmodelcars.com	1	1002	VP Sales
☐ Edit Copy Delete	1076	Firrelli	Jeff	x9273	jfirrelli@classicmodelcars.com	1	1002	VP Marketing
☐ Edit Copy Delete	1088	Patterson	William	x4871	wpatterson@classicmodelcars.com	6	1056	Sales Manager (APAC)
☐ Edit Copy Delete	1102	Bondur	Gerard	x5408	gbondur@classicmodelcars.com	4	1056	Sale Manager (EMEA)
☐ Edit Copy Delete	1143	Bow	Anthony	x5428	abow@classicmodelcars.com	1	1056	Sales Manager (NA)
☐ Edit Copy Delete	1165	Jennings	Leslie	x3291	ljennings@classicmodelcars.com	1	1143	Sales Rep
☐ Edit Copy Delete	1166	Thompson	Leslie	x4065	lthompson@classicmodelcars.com	1	1143	Sales Rep
☐ Edit Copy Delete	1188	Firrelli	Julie	x2173	jfirrelli@classicmodelcars.com	2	1143	Sales Rep
☐ Edit Copy Delete	1216	Patterson	Steve	x4334	spatterson@classicmodelcars.com	2	1143	Sales Rep
☐ Edit Copy Delete	1286	Tseng	Foon Yue	x2248	ftseng@classicmodelcars.com	3	1143	Sales Rep

phpMyAdmin

Current server: MariaDB

Recent Favorites

New

classicmodels

New

customers

employees

offices

orderdetails

orders

payments

productlines

products

footballclub

information_schema

mysql

performance_schema

sys

Source: <https://www.mysqltutorial.org/getting-started-with-mysql/mysql-sample-database>

Server: MariaDB:3307 » Database: classicmodels

Structure SQL Search Query **Export** Import Operations Privileges Routines Events Triggers

Filters

Containing the word:

Table	Action	Rows	Type	Collation	Size	Overhead
☐ customers	★ Browse Structure Search Insert Empty Drop	122	MyISAM	latin1_swedish_ci	20.5 KiB	-
☐ employees	★ Browse Structure Search Insert Empty Drop	23	MyISAM	latin1_swedish_ci	5.7 KiB	-
☐ offices	★ Browse Structure Search Insert Empty Drop	7	MyISAM	latin1_swedish_ci	2.5 KiB	-
☐ orderdetails	★ Browse Structure Search Insert Empty Drop	2,996	MyISAM	latin1_swedish_ci	133.2 KiB	-
☐ orders	★ Browse Structure Search Insert Empty Drop	326	MyISAM	latin1_swedish_ci	27.8 KiB	-
☐ payments	★ Browse Structure Search Insert Empty Drop	273	MyISAM	latin1_swedish_ci	14.4 KiB	-
☐ productlines	★ Browse Structure Search Insert Empty Drop	7	MyISAM	latin1_swedish_ci	5.3 KiB	-
☐ products	★ Browse Structure Search Insert Empty Drop	110	MyISAM	latin1_swedish_ci	33.5 KiB	-
8 tables	Sum	3,864	MyISAM	latin1_swedish_ci	242.9 KiB	0 B

☐ Check all

Exporting tables from "classicmodels" database

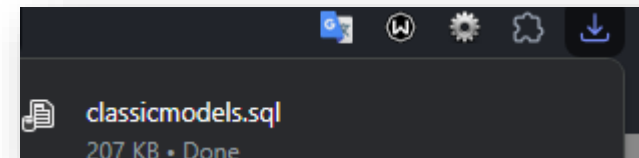
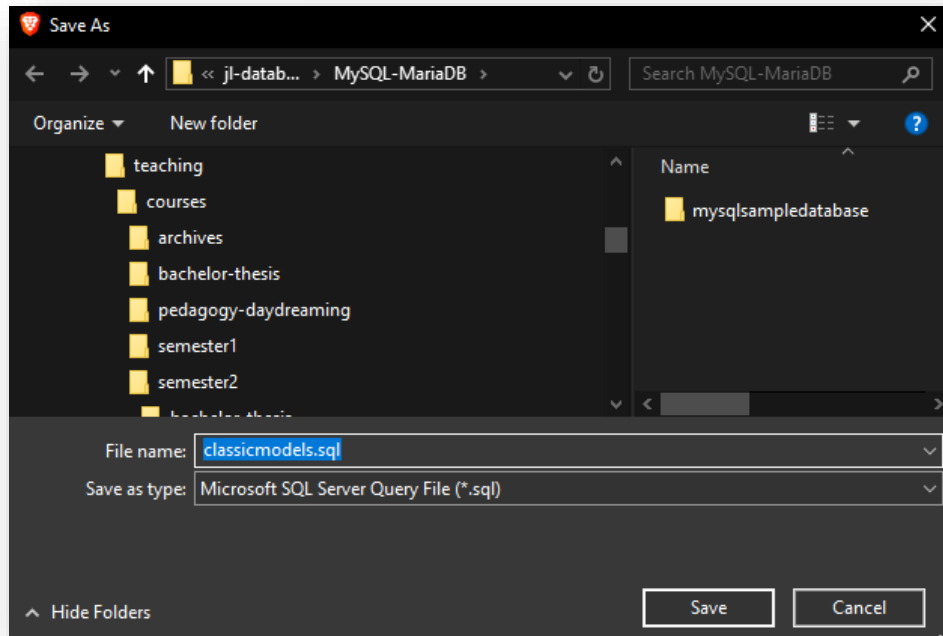
Export method:

- ☒ Quick - display only the minimal options
☐ Custom - display all possible options

Format:

SQL

Export



Exporting tables from "classicmodels" database

Export method:

- ☐ Quick - display only the minimal options
- ☒ Custom - display all possible options

Format:

SQL

Tables:

	Tables	Structure	Data
	Select all	☒	☒
☒	customers	☒	☒
☒	employees	☒	☒
☒	offices	☒	☒
☒	orderdetails	☒	☒
☒	orders	☒	☒
☒	payments	☒	☒
☒	productlines	☒	☒
☒	products	☒	☒

Output:

☐ Rename exported databases/tables/columns

☐ Use LOCK TABLES statement

☐ View output as text

☒ Save output to a file

File name template:  @DATABASE@ ☒ Use this for future exports

Character set of the file: utf-8 

Compression: None 

☐ Export tables as separate files

Skip tables larger than:

The size is measured in MiB.

Format-specific options:

☒ Display comments *(includes info such as export timestamp, PHP version, and server version)*

Additional custom header comment (\n splits lines):

☐ Include a timestamp of when databases were created, last updated, and last checked

☒ Enclose export in a transaction

☐ Disable foreign key checks

☐ Export views as tables

☐ Export metadata

Database system or older MySQL server to maximize output compatibility with:

NONE



Object creation options

Add statements:

☐ Add CREATE DATABASE / USE statement

☒ Add DROP TABLE / VIEW / PROCEDURE / FUNCTION / EVENT / TRIGGER statement

☐ Add CREATE TABLE statement

☒ IF NOT EXISTS (less efficient as indexes will be generated during table creation)

☒ AUTO_INCREMENT value

☒ Add CREATE VIEW statement

☐ Use simple view export

☐ Exclude definition of current user

☐ OR REPLACE view

☒ Add CREATE PROCEDURE / FUNCTION / EVENT statement

☒ Add CREATE TRIGGER statement

☒ Enclose table and column names with backquotes (*Protects column and table names formed with special characters or keywords*)

Data creation options

☐ Truncate table before insert

☐ INSERT DELAYED statements

☐ INSERT IGNORE statements

Function to use when dumping data:

INSERT

Syntax to use when inserting data:

☐ include column names in every INSERT statement

Example: INSERT INTO tbl_name (col_A,col_B,col_C) VALUES (1,2,3)

☐ insert multiple rows in every INSERT statement

Example: INSERT INTO tbl_name VALUES (1,2,3), (4,5,6), (7,8,9)

☒ both of the above

Example: INSERT INTO tbl_name (col_A,col_B,col_C) VALUES (1,2,3), (4,5,6), (7,8,9)

☐ neither of the above

Example: INSERT INTO tbl_name VALUES (1,2,3)

Maximal length of created query

50000

☒ Dump binary columns in hexadecimal notation (for example, "abc" becomes 0x616263)

☒ Dump TIMESTAMP columns in UTC (enables TIMESTAMP columns to be dumped and reloaded between servers in different time zones)

Export

9. Backups in MySQL (and MariaDB)

- MySQL and MariaDB export data in plain text format
- Security issue: you **MUST ALWAYS** **hash the password** in a database!
No password in **CLEAR TEXT**!

ANY QUESTIONS?